# 單元一

# 模組1：測試背景

# 模組 1：知識檢查

Correct answer

**Question 1** 1 / 1 pts

正确还是错误？验证侧重于需求是否满足客户需要，以及需求是否符合客户期望。

True or False? Verification focuses on if the requirements are meeting customer needs and are what the customer wants.

- 正确 True
- ✓ 错误 False

Verification looks at building the product right given the assumption that the requirements are correct. Validation looks at building the right product to bring value and meet the customer's requirements.

Correct answer

**Question 2** 1 / 1 pts

测试人员团队针对一组给定的需求测试软件。需求已由客户批准。测试团队执行的是软件验证还是软件确认操作，或者二者皆不是？

A team of testers are testing software against a given set of requirements. The requirements have already approved by the customer. Is the testing team performing verification or validation of the software, or neither?

- 确认 Validation
- ✓ 验证 Verification

Verification looks at building the product right given the assumption that the requirements are correct. Validation looks at building the right product to bring value and meet the customer's requirements.

- 两者皆不是 Neither

# 模組2：測試完整生命週期

# 模組 2：知識檢查

Correct answer

## Question 1

1 / 1 pts

在敏捷开发中，应多久进行一次测试？

How often does testing happen in agile development?

- 在项目结束时 At the end of the project
- 每月 Monthly
- 每周 Weekly
- ✓ 至少每日一次 At least daily

敏捷测试遵循持续整合的理念，代码被整合和测试至少每天一次。

Agile testing follows the theme of continuous integration where code is integrated and tested at least daily.

Correct answer

## Question 2

1 / 1 pts

在测试驱动开发中，正确的相位顺序是什么？

What is the correct phase order in test driven development?

- 绿、红、重构 Green, Red, Refactor
- ✓ 红、绿、重构 Red, Green, Refactor

红色阶段是输入一小部分关于定义行为的代码测试。接下来，在绿色阶段，输入代码确认上阶段失误的代码无误。在重构阶段，在所有测试成功运作的情况下，提升代码质量。

The red phase is where a small test is written that defines the behavior needed from the code. Then, in the green phase, code is written to ensure the failing test from the previous phase passes. In the refactor phase, the quality of code is improved while all tests are still passing.

- 红、重构、绿 Red, Refactor, Green
- 绿、重构、红 Green, Refactor, Red

Correct answer

## Question 3

1 / 1 pts

系统测试的目标是什么？

What is the objective of system testing?

- 仅测试功能需求 To test only functional requirements
- ✓ 测试功能需求和非功能需求 To test both functional and nonfunctional requirements

> 系统测试是在其他所有组件，例如硬件和软件整合测试（功能需求）以及其他因素例如压力、可用性以及表现测试（非功能需求）被测试后运行。
>
> System testing is when other components, such as hardware and software integration testing (functional requirements) and aspects such as stress, usability, and performance testing (nonfunctional requirements) are tested.

- 测试客户是否接受软件 To test whether the customer accepts the software
- 仅测试非功能需求 To test only nonfunctional requirements

Correct answer

## Question 4

1 / 1 pts

单元层次上的测试的目标是什么？

What is the objective of unit level testing?

- 测试功能需求和非功能需求 To test both functional and nonfunctional requirements
- 测试软件单元是否可以协同工作以执行高级功能和特性 To test whether units of software can work together to perform high level functions and features
- 确认个别开发人员的单元是否经过正确测试 To validate that an individual developer's unit is tested properly
- ✓ 验证个别开发人员的单元是否经过正确测试 To verify that an individual developer's unit is tested properly

> 单元级别测试是每一个开发人员测试他们各自单元代码以确保这些单元都运作正常。
>
> Unit level testing is where each developer tests their unit (section of code) to ensure it is working properly

# 模組3：測試最佳實踐和標準

# 模組 3：知識檢查

Correct answer

## Question 1

1 / 1 pts

系统测试应由哪组用户执行？

Which group of people should perform system testing?

- 整合团队 An integration team
- ✓ 独立测试团队 An independent test team

  A testing attitude is to gain independence to counter bias during testing.

- 负责开发和测试工作的人员的组合 A combination of developers and testers
- 开发人员 Developers

Correct answer

## Question 2

1 / 1 pts

系统测试的主要目标是什么？

What is(are) the primary objective(s) of system testing?

- 预测可靠性 To predict reliability
- ✓ 找到重要问题和预测可靠性 To find important problems and predict reliability

  Finding important problems that affect the customer greatly is more important than finding many minor problems in system testing. Testing must also provide an estimate of reliability.

- 对所有问题执行穷尽搜索 To perform an exhaustive search for all the problems
- 找到所需数量的问题 To find a required number of problems

Correct answer

## Question 3

1 / 1 pts

何时停止测试最合理？

When is a reasonable time to stop testing?

- 证明满足至少一个需求 Once it is demonstrated that at least one requirement is met
- ✓ 已达到测试目标 Once we have met our test objectives.

  Test objectives typically include testing all of the functional and nonfunctional requirements.
- 没有需要测试的需求 No testing of requirements is needed.
- 证明已能满足重要需求 Once it is demonstrated that the important requirements are met

Correct answer

## Question 4

1 / 1 pts

为什么缺陷会群集？

Why do defects cluster?

- 缺陷实际上均匀地分布在每一千行代码中 Defects are actually distributed evenly across every thousand lines of code.
- 因为基本代码没有任何变化 Because there are no changes happening to the codebase
- 因为开发人员故意包含它们，以供测试人员发现 Because a developer intentionally includes them to be found by testers
- ✓ 由于代码复杂性、程序员技能等 Because of the complexity of code, programmer skill, etc.

  Defect clustering happens for many reasons. Some parts of the code are more complex, causing more defects.

# 單元 1：測驗

Correct answer

**Question 1** **1 / 1 pts**

测试仅表明______

Testing only shows the ____

- 正确性 Correctness
- ✓ 存在缺陷 Presence of defects
- 存在缺陷和正确性 Presence of defects and correctness

Correct answer

**Question 2** **1 / 1 pts**

正确还是错误？基于模型的测试需要对系统的行为进行建模。

True or False? Model based testing requires modeling the behavior of the system.

- ✓ 正确 True
- 错误 False

Correct answer

## Question 3

1 / 1 pts

打破它（break it）测试属于哪种活动？

What kind of activity is “break it” testing?

- 确认和验证活动 Validation and verification activity
- 验证活动 Verification activity
- 确认和验证皆不是 Neither a validation nor verification activity
- ✓ 确认活动 Validation activity

Validation is to validate functional and non functional features. "Break it" is an activity to test the reliability or robust  of the software.

Correct answer

## Question 4

1 / 1 pts

哪种情况不属于测试中的典型错误？

What is not considered a classic testing mistake?

- 不对文档进行测试 Not testing the documentation
- 开始的太晚 Starting too late
- ✓ 给测试团队配备领域专家 Staffing the team with domain experts
- 不关注使用性问题 Not focusing on usability issues

Correct answer

## Question 5

1 / 1 pts

正确还是错误？功能测试着眼于性能需求、使用性等。

True or False? Functional testing looks at performance requirements, usability, etc.

- 正确 True
- ✓ 错误 False

Correct answer

## Question 6

1 / 1 pts

一个项目的截止日期是在两天后，但是测试人员需要至少三天时间才能完成测试。根据国际软件测试资格委员会（ISTQB）道德规范，测试人员团队应该做什么？

A project is due in two days, and the team of testers needs at least three days to finish testing. Based on the ISTQB Code of Ethics, what should the team of testers do?

- 只测试到截止日期 Only test until the deadline
- 立刻停止测试 Stop testing immediately
- ✓ 请求延长期限，以保证软件能被全面测试 Ask for an extension on the deadline to be able to test the software fully

Correct answer

## Question 7

1 / 1 pts

正确还是错误？测试是通过执行所有可能的数据集来检查程序的行为。

True or False? Testing is the examination of the behavior of the program by executing it on all of the possible data sets.

- 正确 True
- 错误 False

Wrong answer

## Question 8

0 / 1 pts

在测试驱动开发中的哪一个步骤等价于软件开发过程中收集需求的步骤？

In test driven development, what is the equivalent of gathering requirements during the software development process?

- 收集测试人员的输入 Gathering testers' input
- 收集测试目标 Gathering test objectives
- 收集样本测试用例 Gathering sample test cases
- 收集可能的测试设计 Gathering possible test designs

Correct answer

## Question 9

1 / 1 pts

正确还是错误？认证软件测试人员应确保他们交付的软件符合现有的最高专业标准。

True or False? Certified software testers should ensure that the deliverables they provide meet the highest professional standards possible.

- ✓ 正确 True
- 错误 False

Correct answer

## Question 10

1 / 1 pts

正确还是错误？可靠性是指一个软件程序（系统）在给定的时间段内运行，不出现软件错误的概率。

True or False? Reliability is the probability that a software program (system) operates for some given time period without software error.

- ✓ 正确 True
- 错误 False

# 單元二

# 模組1：基於場景的測試

# 模組 1：知識檢查

Correct answer

## Question 1

1 / 1 pts

基于需求的测试是哪一种活动？

What kind of activity is requirement-based testing?

- 基于需求的测试既不是验证，也不是确认活动。Requirements based testing is neither a verification or validation activity.
- 既是验证活动，也是确认活动。Primarily both a verification and validation activity
- 主要是一种确认活动 Primarily a validation activity
- ✓ 主要是一种验证活动 Primarily a verification activity

测试用例是根据需求清单开发的，在基于需求的测试中用于验证是否已满足每项需求。这是一种验证活动。当中没有核对是否有正确的需求(这是一种确认活动)。Test cases are developed around a list of requirements to verify that each requirement is met in requirements based testing. This is a verification activity. There is no check as to whether there are the right requirements, which is a validation activity.

Correct answer

## Question 2

1 / 1 pts

基于场景的测试是哪一种活动？

What kind of activity is scenario-based testing?

- 主要是一种确认活动。Primarily a validation activity
- 基于需求的测试既不是验证，也不是确认活动。Requirements based testing is neither a verification or validation activity.
- ✓ 既是验证，也是确认活动。Primarily both a verification and validation activity
- 主要是一种验证活动。Primarily a verification activity

在基于场景的测试中，需求是通过创建用使用例开发的，然后再围绕着使用用例开发测试。通过使用用例能更容易地确认一种活动是否有正确的需求，然后再验证每项需求是否都有测试用例，类似于课程中的打包示例。In scenario based testing, requirements are developed by creating use cases, and then tests are developed around the use cases. Having use cases makes it easier to validate whether the right requirements are there for an activity and then verify that test cases exist for each requirement, similar to the packing example from the lecture.

Correct answer

## Question 3

1 / 1 pts

下列哪一步不属于使用用例构建中的步骤？

What is not a step in use case construction?

- 确定行为者（确定系统范围）。Identify the actors (determine the scope of the system)
- 详细描述各使用用例，将每个处理流程确认为一个场景。Detail each use case identifying each of its processing flows as a scenario
- 确定行为者如何使用系统完成功能。Identify how an actor uses the system to accomplish its functions
- ✓ 确定如何测试各使用用例。Identify how each use case can be tested

使用用例构建包含行为者，行为者如何使用系统以及针对每种行为的使用用例。从测试的角度来看，确定如何测试各使用用例是之后的步骤。Use case construction includes the actors, how the actors use the system and use cases for each behavior. Identifying how each use case can be tested comes afterwards when looking from a testing perspective.

# 模組2_講座一：等價類劃分測試

# 模組 2_講座一：知識檢查

Correct answer

## Question 1

1 / 1 pts

正确还是错误？等价类划分是在具有多个独立的输入时采用的良好技术。

True or False?

Equivalence partitioning is a good technique to utilize when there are multiple independent inputs.

- 错误 False
- ✓ 正确 True

等价类划分必须应用于独立的输入。在本课程的 abs(x) 示例中，必须测试负值、0 和正值，它们都是独立的输入。Equivalence partitioning must be applied with independent inputs. In the lecture example of abs(x), it is necessary to test a negative value, 0, and a positive value which are all independent inputs.

Correct answer

## Question 2

1 / 1 pts

哪一项不是等价类划分步骤？

What is NOT an equivalence partitioning step?

- ✓ 编写多个测试用例，尽可能多的覆盖尚未覆盖的无效等价类分区。Write test cases covering as many of the uncovered invalid equivalence partitions as possible.

一次只能有一个测试用例去覆盖每个未覆盖的无效等价类。因为等价类划分用于黑盒测试过程，因此如果测试了多个无效类，则无法判断哪个无效类导致了错误。There must only be one test case that covers each uncovered invalid equivalence partition at a time. Since equivalence partitioning is used during black box testing, if there are multiple invalid partitions tested, it is not known which invalid partition caused the error.

- 对于每个无效等价类，编写一个测试用例去覆盖其中一个尚未覆盖等价类。For each invalid equivalence partition, write a test case that covers one of the uncovered equivalence partitions.
- 对于每个输入，识别一组等价类并进行标记。For each input, identify a set of equivalence partitions and label them.
- 编写多个测试用例，尽可能多的覆盖尚未覆盖的有效等价类。Write test cases covering as many of the uncovered valid equivalence partitions as possible.

Correct answer

## Question 3

1 / 1 pts

测试一个密码长度必须介于 6-8 个字符之间的密码程序，一组良好的等价类是什么？

What are a good set of equivalence partitions for a password testing program where a password must be between 6-8 characters?

1. 6-8 个字符；2. < 6 个字符；3. > 8 个字符（1. 6-8; 2. < 6 characters; 3. > 8 characters）

这些等价类测试所有可能的输入，可能是密码：密码长度介于 6-8 个字符之间的有效密码，密码< 6 个字符和 > 8 个字符的都是无效密码。

These equivalence partitions test all the possible inputs there could be for a password: a valid password between 6-8 characters, an invalid password with < 6 characters, and an invalid password with > 8 characters.

1. 6-8 个字符；2. < 6 个字符（1. 6-8 characters; 2 < 6 characters）

1. < 6 个字符；2. > 8 个字符（1. < 6 character; 2. > 8 characters）

1. 6-8 个字符（1. 6-8 characters）

# 模組2_講座二：邊界值測試

# 模組 2_講座二：知識檢查

Correct answer

**Question 1** 1 / 1 pts

有效的学生 ID 只能包含 2 （含）到 7（含）之间的数字。需要测试的正确边界值是什么？

A valid student ID can only contain digits between and including 2 and 7. What are the correct boundary values that need to be tested?

- 2,7
- 1,8
- 1,2
- ✓ 1,2,7,8

1 和 2 测试第一个边界的有效和无效值 (大于或包含 2 的数字)。7 和 8 测试第二个边界的有效和无效值。

1 and 2 test valid and invalid values for the first boundary of a digit greater than or including 2. 7 and 8 test valid and invalid values for the second boundary.

Correct answer

**Question 2** 1 / 1 pts

进行边界值分析时，需要选择仅位于输入和输出等价类划分边界或其上的测试用例。

Boundary value analysis requires selecting test cases that are only on or above the edges of the input and output equivalence partitions.

- 正确 True
- ✓ 错误 False

进行边界值分析时，需要选择位于划分边界、其上和其下的测试用例。

BV analysis requires selecting test cases that are on, above, and below the edges of the partitions.

# 模組2_講座三：因果分析和問題示例

# 模組 2_講座三：知識檢查

Correct answer

**Question 1** 1 / 1 pts

给定 2 个输入变量（年龄和身高）、维生素的输出、以及一个根据年龄和身高计算一个人应该服用的维生素数量的函数，可以使用哪种方法来对这个示例进行测试?

Given 2 input variables, age and height, an output of vitamins, and a function that computes the number of vitamins a person should take based on age and height, which method(s) can be used to test this example?

- 因果分析和等价类划分 Both cause effect analysis and equivalence partitioning
- 等价类划分 Equivalence partitioning
- ✓ 因果分析 Cause Effect Analysis

  当输入相互依赖时，可以使用因果分析。当输入相互独立时，才可以使用等价类划分。Cause effect analysis can be used when the inputs are dependent on each other. Equivalence partitioning is only used when inputs are independent of each other.

- 因果分析和等价类划分均不可用 Neither cause effect analysis nor equivalence partitioning

Correct answer

**Question 2** 1 / 1 pts

减少因果决策表中测试用例数量的一种方法是什么?

What is one way to reduce the number of test cases in a cause and effect decision table?

- 仅编写有效的测试用例 Write only valid test cases
- ✓ 作出划分是如何相关的假设 Make assumptions about how the partitions are related

  从课堂上的例子中，在客户和订单之间的错误处理方式上作出假设，可以减少所需测试用例的数量。From the example in lecture, making assumptions about how error handling between a customer and order happens allows us to reduce the number of test cases needed.

- 根据划分的数量编写测试用例 Write as many test cases as there are partitions

# 模組3_講座一：測試非同步事件

# 模組 3_講座一：知識檢查

Correct answer

**Question 1** 1 / 1 pts

创建测试同步事件的时间线是有助于验证还是确认？

Does creating a timeline to test synchronous events help with verification or validation?

- ✓ 确认 Validation

  在时间线中添加事件有助于识别遗漏的需求，这是确认所必需的。Adding in events in a timeline helps identify missing requirements, which is necessary for validation.

- 验证 Verification

Correct answer

**Question 2** 1 / 1 pts

时间线对应于一组与时间相对应的使用用例。

A timeline corresponds to a set of use cases mapped against time.

- 正确 True
- ✓ 错误 False

  每个使用用例都需要有各自的时间线。Each use case requires its own timeline.

# 模組3_講座二：基於狀態的測試和問題示例

# 模組 3_講座二：知識檢查

Correct answer

**Question 1** 1 / 1 pts

正确还是错误？测试状态机的基本策略是访问每个状态并测试每个转移。

True or False? The basic strategy of testing a state machine is to visit each state and test each transition.

✓ 正确 True

测试每个状态/事件组合可提供最小状态机覆盖。
Testing each state / event combination provides a minimal state machine cover.

错误False

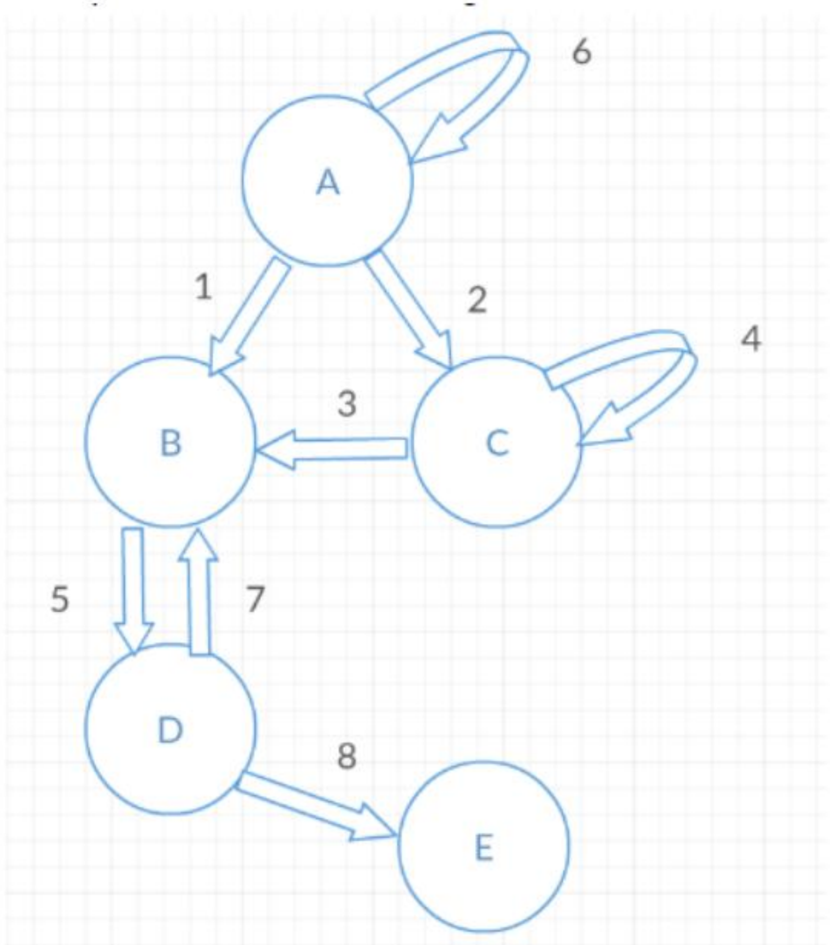
Correct answer

## Question 2

1 / 1 pts

根据下面的状态图，正确的测试序列是什么？

Given the state diagram below, what is a correct test sequence?




- [x] 1 5 8

  初始转移是 1，然后通过转移 5 和 8，到达终止状态。The starting transition is 1, and then through transition 5 and 8, we reach the end state.

- [ ] 1 5 7 8
- [ ] 3 5 8
- [ ] 6 2 1

# 模組4：基於模型的測試

# 模組 4：知識檢查

Correct answer

**Question 1** 1 / 1 pts

以下哪项不被认为是基于模型的测试的优点？

What is NOT considered to be an advantage of model-based testing?

- 建模需要手动生成测试用例。Modeling requires manual test case generation.

  建模为自动化生成测试提供了可能性，不需要手动生成测试用例。Modeling gives potential for automated test generation, and does not require manual test case generation.

- 建模为自动化生成测试提供了可能性。Modeling gives potential for automated test generation.
- 建模允许行为变更直接转换为测试变更。Modeling allows for changes in behavior to be directly translated into test changes.
- 建模可提供精确性并减少歧义。Modeling provides precision and reduces ambiguity.

Correct answer

**Question 2** 1 / 1 pts

基于模型的测试步骤的正确顺序是什么？

What is the correct order of steps for model-based testing?

- 选择测试生成准则，创建系统模型，生成测试，执行测试。Select test generation criteria, create a system model, generate tests, execute tests
- 创建系统模型，选择测试生成准则，生成测试，执行测试。Create a system model, select test generation criteria, generate tests, execute tests

  此内容源于课程幻灯片。This is from the lecture slides.

- 生成测试，创建系统模型，选择测试生成准则，执行测试。Generate tests, create a system model, select test generation criteria, execute tests

# 單元 2：測驗

Correct answer

**Question 2** 1 / 1 pts

正确还是错误？决策树和决策表总是会产生相同数量的测试用例。

True or False? Decision trees and decision tables will always result in the same number of test cases.

- 正确 True
- ✓ 错误 False

Correct answer

**Question 1** 1 / 1 pts

三个输入变量（产品编号、价格和评级）用于确定产品是否属于"最受用户欢迎"类别。评级与价格有关，产品编号与其他输入无关。哪种方法可用于测试此问题？

Three input variables, product number, price, and ratings are used to determine whether a product belongs in the "People's Choice" category. Ratings is dependent on price and product number is independent of the other inputs. What method can be used to test this problem?

- 等价类划分 Equivalence partitioning
- 时间线 Timelines
- 基于状态的测试 State Based Testing
- ✓ 等价类划分 x 因果分析 Equivalence partitioning x Cause Effect Analysis

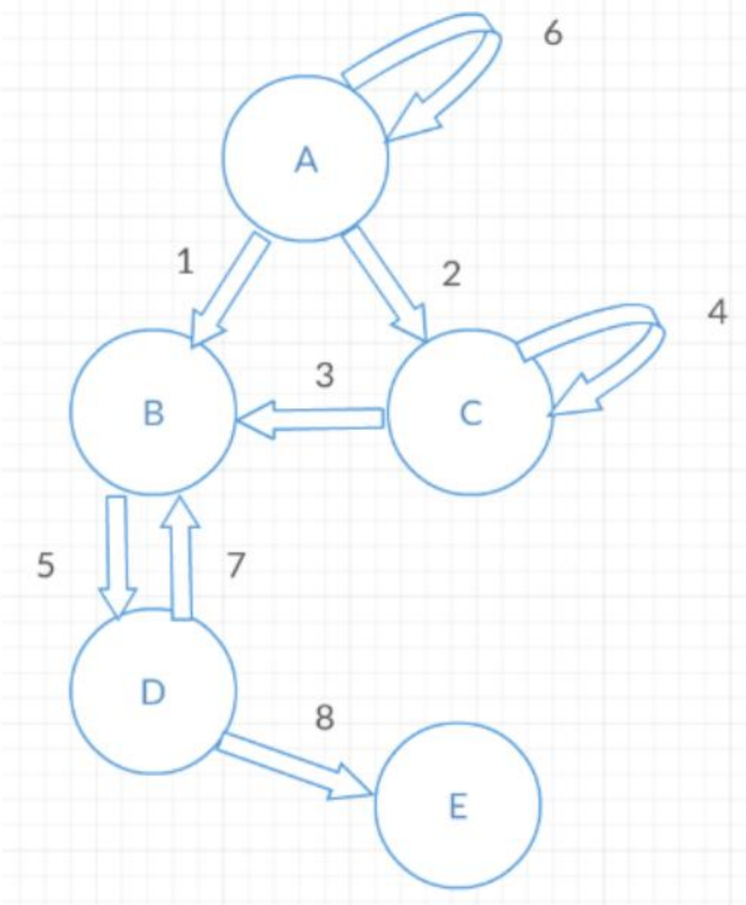
Correct answer

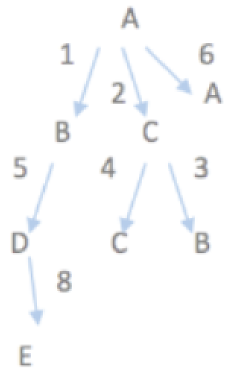
## Question 3

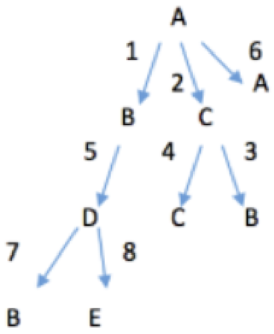
1 / 1 pts

根据下面的状态机图，正确的状态测试树是哪个？

Given the state machine diagram below, what is the correct state testing tree?



Correct answer

## Question 4

1 / 1 pts

根据以下要求，选择用于测试问题的适当决策表。

根据车主类型（教师、职工、学生）和停车位置（车库或停车场）计算通行证的停车费。教师只能在车库停车，需支付 200 美元。学生选择一个车库车位需支付 100 美元，选择一个停车场车位需支付 50 美元。职工可以在车库或停车场停车，两种选择都需支付 200 美元。

请选择正确的决策表。

Given the following requirements, select the appropriate decision table for testing the problem.

Parking rates for a permit calculated based on the type of vehicle owner (faculty, staff, student) and the location of the parking (garage or lot). Faculty may park in the garage only at $200. Students pay $100 for a garage spot and $50 for a lot spot. Staff may park either in the garage or the lot for $200.

Please select the correct decision table.

| | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|
| 教师 Faculty | X | X | | | | |
| 职工 Staff | | | | | X | X |
| 学生 Student | | | X | X | | |
| | | | | | | |
| 车库 Garage | X | | X | | X | |
| 停车场 Lot | | X | | X | | X |
| | | | | | | |
| $200 | X | | | | X | X |
| $100 | | | X | | | |
| $50 | | | | X | | |
| 不允许 Not Allowed | | X | | | | |

Correct answer

## Question 5

1 / 1 pts

正确还是错误？基于场景的测试有助于识别遗漏的需求。

True or False? Scenario based testing helps identify missing requirements.

- 错误 False
- 正确 True

Correct answer

## Question 6

1 / 1 pts

正确还是错误？决策树是执行因果分析的一种方法。

True or False? Decision trees are a way to perform cause and effect analysis.

- 错误 False
- 正确 True

Correct answer

## Question 7

1 / 1 pts

在一次考试中，学生最少可以获得 20 分，最多可以获得 50 分。对分数的有效等价类划分是什么？

A student can score a minimum of 20 points and a maximum of 50 points on an exam. What is a valid equivalence partition of scores?

- 小于 20 以及大于 50 （less than 20 and greater than 50）
- ✓ 20 到 50 （20 to 50）
- 大于 50 （greater than 50）
- 小于 20 （less than 20）

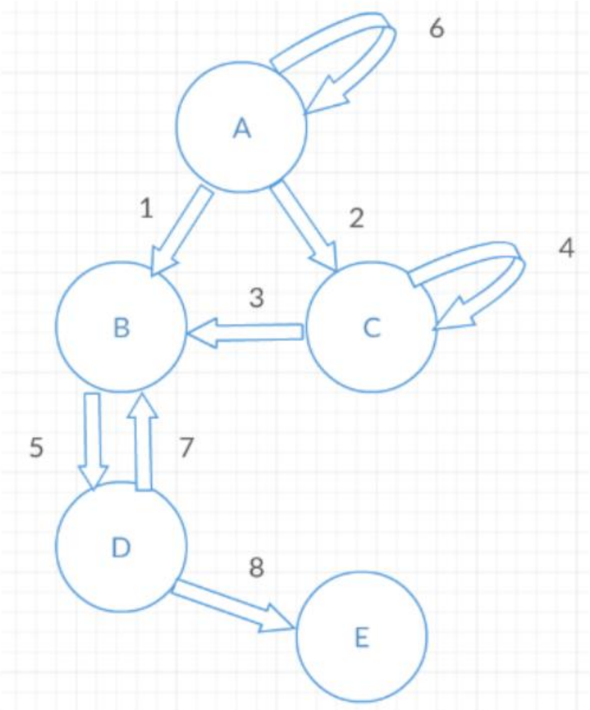
Correct answer

## Question 8

1 / 1 pts

从下面的状态测试图中，正确的测试序列数量是多少？

From the state testing diagram below, what is the correct number of test sequences?




- 6
- 4
- ✓ 5
- 3

The five test sequences are:

6

1, 5, 7

1, 5, 8

2, 4

2, 3

We do not need to test the states after 3--> B, as they have been tested by 1-->B

Correct answer

## Question 9

1 / 1 pts

以下哪项不被视为状态机中的错误？

Which of the following is not considered as an error in a state machine?

- 死状态 Dead state
- 无法到达状态 Unreachable state
- ✓ 来自同一状态的两个不同事件 Two different events emerging from the same state

  在时间轴中添加事件有助于识别缺少的需求，这对于验证是必需的。Adding in events in a timeline helps identify missing requirements, which is necessary for validation.
- 来自同一状态的两个相同事件 Two identical events emerging from the same state

Correct answer

## Question 10

1 / 1 pts

表格中的文本框只能接受 19 到 26 范围内的数值。请识别无效的等价类划分。

A text field on a form only accepts numeric values in the range of 19 to 26. Please identify the invalid equivalence partitions.

- ✓ 小于 19 以及大于 26 （less than 19 and greater than 26）
- 19 到 26 （19 to 26）
- 小于 19 以及 19 到 26 （less than 19 and 19 to 26）
- 19 到 26 以及大于 26 （19 to 26 and greater than 26）

# 單元三

# 模組1：組合測試技術

# 模組 1：知識檢查

Correct answer

## Question 1

1 / 1 pts

（正确或错误）实验设计成对组合需要系统测试输入的所有组合。

(True or False) Design of Experiments pairwise combination involves systematically testing all combinations of inputs.

错误 False

只有每次输入的成对数值才一起测试，而非输入的所有数值组合。

Only pairs of values for each input are tested together, not all combinations of values of inputs.

正确 True

Correct answer

## Question 2

1 / 1 pts

（正确或错误）组合覆盖关注的是单独测试各参数数值。

(True or False) Combinatorial coverage looks at parameter values being individually tested.

正确 True

错误 False

组合覆盖关注的是如何一起测试参数数值组合，而非单独的参数。

Combinatorial coverage looks at how combinations of parameter values are tested together

Correct answer

## Question 3

1 / 1 pts

假设我们在测试一个有 3 个变量的功能函数：

变量 A：值为 0 和 1

变量 B：值为 0 和 1

变量 C：值为 0 和 1

以下测试实现的 2 路变量值配置的总覆盖率是多少：

A=0; B=0; C=0

A=0; B=1; C=1

A=1, B=1, C=0

Assume we are testing a function with 3 variables:

Variable A: has values 0 and 1

Variable B: has values 0 and 1

Variable C: has values 0 and 1

What is the total 2-way variable value configuration coverage achieved by the following tests:

A=0; B=0; C=0

A=0; B=1; C=1

A=1, B=1, C=0

- 10/12
- 8/12
- 7/12
- ✓ 9/12

可能组合：A=0 B=0; A=0 C=0; B=0 C=0; A=0 B=1; A=0 C=1; B=1 C=1; A=1 B=1; A=1 C=0; B=1 C=0

Possible combinations: A=0 B=0; A=0 C=0; B=0 C=0; A=0 B=1; A=0 C=1; B=1 C=1; A=1 B=1; A=1 C=0; B=1 C=0

Correct answer

## Question 4

1 / 1 pts

实验设计的目标是什么？

What is the goal of the design of experiments?

- 任意数量的组合。 Test a random number of combinations
- ✓ 尽量减少我们需要运行的测试次数 Minimize the number of tests we need to run

  我们测试每次输入的成对数值，以尽量减少测试的次数。

  We are testing pairs of values for each input to minimize the number of tests.

- 尽量增加我们需要运行的测试次数。Maximize the number of tests we need to run

Correct answer

## Question 5

1 / 1 pts

给定 3 个输入：P1 取值为 V1 和 V2，P2 取值为 V3、V4 和 V5，P3 取值为 V6和 V7，结对组合实验设计的正确测试是？

Given 3 inputs: P1 with values V1 and V2; P2 with values V3, V4, and V5 and P3 with values V6 and V7, what are the correct tests for a pairwise combination design of experiments?

| | | |
|---|---|---|
| V3 | V1 | V6 |
| V4 | V1 | V6 |
| V5 | V1 | V6 |

| | | |
|---|---|---|
| V3 | V1 | V6 |
| V3 | V2 | V7 |
| V4 | V1 | V6 |
| V4 | V2 | V7 |
| V5 | V1 | V6 |
| V5 | V2 | V7 |

| | | |
|---|---|---|
| V3 | V1 | V6 |
| V3 | V2 | V7 |
| V4 | V1 | V7 |
| V4 | V2 | V6 |

✓

| | | |
|---|---|---|
| V3 | V1 | V6 |
| V3 | V2 | V7 |
| V4 | V1 | V7 |
| V4 | V2 | V6 |
| V5 | V1 | V6 |
| V5 | V2 | V7 |

本表测试了每种组合的成对数值。

This table tests every combination of pairs of values.

# 模組2：變異測試

# 模組 2：知識檢查

Correct answer

## Question 1

1 / 1 pts

变异测试是为了将缺陷引入被测试的程序中，以检测测试用例是否能发现变异体。

Mutation testing is designed to introduce defects into a program undergoing testing to see if the test cases can detect the mutant.

- 正确 True

  这是变异测试的定义。

  This is the definition of mutation testing.

- 错误 False

Correct answer

## Question 2

1 / 1 pts

下列哪项不属于变异的示例？

What is NOT an example of a mutation?

- 重新命名一个类模块，以及调用或发现该类模块的任何地方 Renaming a class and anywhere that the class is called or found

  如果一个类模块在所有地方都被重新命名，就无法在程序中引入错误（变异体）。

  If a class is renamed everywhere, that would not introduce an error (mutant) in the program.

- 修改算术运算 Modifying arithmetic operations
- 修改布尔表达式 Modifying Boolean expressions
- 修改变量 Modifying variables

# 模組3：模糊測試

# 模組 3：知識檢查

Correct answer

**Question 1** 0 / 0 pts

模糊测试包括自动创建的随机、无效或非预期的输入。

Fuzz testing consists of random, invalid or unexpected inputs that are created automatically.

正确 True

模糊测试是一种方法，这一方法将自动生成无效、随机或非预期的输入。

Fuzz testing is an approach to testing where invalid, random or unexpected inputs are automatically generated.

错误 False

Correct answer

**Question 2** 0 / 0 pts

（正确还是错误）模糊测试仅查找不良行为或崩溃。

(True or False) Fuzz testing looks only for undesirable behavior or crashes.

正确 True

模糊测试并非着眼于特定的输入或输出，而是查找错误或过失的行为。

Fuzz testing is not looking at specific inputs or outputs, but is instead looking for an error or a wrong behavior.

错误 False

# 模組4：蛻變測試

# 模組 4：知識檢查

Correct answer

## Question 1

1 / 1 pts

（正确还是错误）蜕变测试假设：如果某个程序具有输入 x，并且产生输出 y，当对输入x 作出更改时，同一更改不会反映在输出 y 中。

(True or False) Metamorphic testing makes the assumption that if there is a program with input x that results in output y, and there is a change to input x, that same change is not reflected in output y.

- 正确 True
- ✓ 错误 False

蜕变测试假设：当对输入作出更改时，可以预测输出也会发生更改。Metamorphic testing makes the assumption that when changes are made to an input, it is possible to predict changes on the output.

Correct answer

## Question 2

1 / 1 pts

如果不使用计算器，在此示例中使用蜕变测试时，第三个测试用例的预期输出是什么？

初始测试：5 10 15 20 25 标准偏差结果：7.2

第二个测试：5 15 25 35 45 标准偏差结果：14.4

第三个测试：15 20 25 30 35 标准偏差结果：___

Without using a calculator, what would be the expected output of this example using metamorphic testing for the third test case?:

Initial Test: 5 10 15 20 25 Stan. Dev Result: 7.2

Second Test: 5 15 25 35 45 Stan Dev Result: 14.4

Third test: 15 20 25 30 35 Stan Dev Result: ___

- 3.6
- 28.8
- ✓ 7.2

当值增加 5 时，标准偏差为 7.2，在第三个测试中，值增加了 5。When the values are incremented by 5, the standard deviation was 7.2 In the third test, the values are incremented by 5.

- 14.4

# 模組5：基於缺陷的測試

# 模組 5：知識檢查

Correct answer

**Question 1** 1 / 1 pts

（正确或错误）基于缺陷的测试只能应用于单元层次。

(True or False) Defect-based testing can only be applied at the unit level.

- 错误 False ✓

  基于缺陷的测试可以应用于任何层次的测试。

  Defect based testing can be applied at any level of testing.

- 正确 True

Correct answer

**Question 2** 1 / 1 pts

（正确或错误）基于缺陷的测试着眼于创建针对特定缺陷类别的测试用例。

(True or False) Defect based testing looks to create test cases that target specific defect categories.

- True ✓

  基于缺陷的测试可以针对 Beizer 通用缺陷分类表中的任何缺陷类别。

  Defect based testing can target any defect category from the Beizer Generic Defect Taxonomy Categories.

- False

# 模組6：探索性測試

# 模組 6：知識檢查

Correct answer

**Question 1** 1 / 1 pts

在探索性测试中，所有测试脚本都无需预先开发。

In exploratory testing, all test scripts are not developed in advance.

错误 False

✓ 正确 True

接下来要测试的项目是基于之前测试的结果。

What is tested next is based on the results of the previous tests.

Correct answer

**Question 2** 1 / 1 pts

探索性测试关注可帮助检测特定错误的旅程。

Exploratory testing focuses on a tour that helps detect a specific error.

✓ 错误 False

正确。探索性测试可能包括关注不同错误的需求、功能、持续使用、文档等过程。A explanation: Exploratory testing can consist of requirements, features, continuous use, documentation, etc. tours that focus on different errors.

正确 True

# 單元 3：測驗

Correct answer

**Question 1** 1 / 1 pts

（正确或错误）基于变异模糊的测试是基于一个测试“种子”，也称为有效测试用例。

(True or False) Mutation fuzz based testing is based upon a testing “seed”, otherwise known as a valid test case.

- ✓ 正确 True
- 错误 False

Correct answer

**Question 2** 1 / 1 pts

基于缺陷的测试利用缺陷分类法来创建测试用例，以针对特定的缺陷。

Defect-based testing utilized defect taxonomies to create test cases to target certain defects.

- ✓ 正确 True
- 错误 False

Correct answer

## Question 3

1 / 1 pts

探索性测试包括基于以往的测试经历同时设计和创造测试。

Exploratory testing consists of designing and creating tests concurrently based on past testing experiences.

- 错误 False
- ✓ 正确 True

Correct answer

## Question 4

1 / 1 pts

给定 3 个输入：P1 取值为 V1 和 V2；P2 取值为 V3，P3 取值为 V4 和 V5，结对组合实验设计的正确测试是？

Given 3 inputs: P1 with values V1 and V2; P2 with value V3, and P3 with values V4 and V5, what are the correct tests for a pairwise combination design of experiments?

- 

| | | |
|---|---|---|
| V1 | V4 | V3 |
| V1 | V5 | V3 |
| V2 | V4 | V3 |

- 

| | | |
|---|---|---|
| V1 | V3 | V4 |
| V2 | V3 | V5 |

- ✓

| | | |
|---|---|---|
| V1 | V4 | V3 |
| V1 | V5 | V3 |
| V2 | V4 | V3 |
| V2 | V5 | V3 |

- 

| | | |
|---|---|---|
| V1 | V3 | V5 |
| V2 | V3 | V4 |

Correct answer

## Question 5

1 / 1 pts

程序 x 的所有测试用例都通过了，但是程序 x 的变异体 y 的一些测试用例失败了。初始测试用例是良好的测试用例。

All test cases pass for program x, but some test cases fail for a mutant y of program x. The original test cases are good test cases.

- 错误 False
- ✓ 正确 True

Correct answer

## Question 6

1 / 1 pts

假设我们在测试一个有 3 个变量的功能函数：

变量 A：值为 0 和 1

变量 B：值为 0 和 1

变量 C：值为 0 和 1

通过以下测试实现 3 路变量值配置的总覆盖率是多少：

A=0; B=0; C=0

A=0; B=1; C=1

A=1, B=1, C=0

A=1, B=1, C=1

Assume we are testing a function with 3 variables:

Variable A: has values 0 and 1

Variable B: has values 0 and 1

Variable C: has values 0 and 1

What is the total 3-way variable value configuration coverage achieved by the following tests:

A=0; B=0; C=0

A=0; B=1; C=1

A=1, B=1, C=0

A=1, B=1, C=1

- ✓ 4/8
- 3/8
- 4/12
- 6/12

Correct answer

## Question 7

1 / 1 pts

提供 3 个输入：P1 取值为 V1、V2、V3；P2 取值为 V4、V5、V6，P3 取值为 V7 和 V8，成对组合试验设计有多少个测试？

Given 3 inputs: P1 with values V1, V2, V3; P2 with values V4, V5, V6 and P3 with values V7 and V8, how many tests are there for a pairwise combination design of experiments?

- ✓ 9
- 6
- 18
- 8

Correct answer

## Question 8

1 / 1 pts

（正确或错误）实验设计允许对单一和组合输入进行检查。

(True or False) Design of Experiments allows for examination of both single and combinations of inputs.

- ✓ 正确 True
- 错误 False

Correct answer

## Question 9

1 / 1 pts

假设我们在测试一个有 3 个变量的功能函数：

变量 A：值为 0 和 1

变量 B：值为 0 和 1

变量 C：值为 0 和 1

以下测试实现的 2 路变量值配置的总覆盖率是多少：

A=0; B=0; C=1

A=0; B=1; C=1

A=1, B=0, C=0

Assume we are testing a function with 3 variables:

Variable A: has values 0 and 1

Variable B: has values 0 and 1

Variable C: has values 0 and 1

What is the total 2-way variable value configurati
following tests:

A=0; B=0; C=1

A=0; B=1; C=1

A=1, B=0, C=0

- 3/8
- 7/12
- 9/12
- ✓ 8/12

Correct answer

## Question 10

1 / 1 pts

（正确或错误）组合覆盖和变异测试是评估测试质量的方法。

(True or False) Combinatorial coverage and mutation testing are ways to assess test quality.

- 错误 False
- ✓ 正确 True

# 單元四

# 模組1_講座1：控制流測試

# 模組 1_講座1：知識檢查

Correct answer

**Question 1** **1 / 1 pts**

根据下方给定的代码，需要多少测试用例才能达到 100% 复合条件覆盖？

Given the code below, how many test cases are needed to achieve 100% multiple condition coverage?

If a < 12 and b = 5 or c > 15

X = 50;

Else

X = 25;

- 2
- 3
- ✓ 8

须测试 A < 12 时的正确/错误，须测试 b = 5 时的正确/错误，以及测试 c > 15 时的正确/错误，且每种组合都要互相测试。(2x2x2 = 8)

A < 12 needs to be tested as T/F, b = 5 needs to be tested as T/F, and c > 15 needs to be tested as T / F, and every combination must be tested with each other. (2x2x2 = 8)

- 6

Correct answer

**Question 2** **1 / 1 pts**

正确还是错误？

彻底执行所有需求，可保证代码的完全覆盖。

True or False?

Thoroughly executing all requirements guarantees full code coverage.

- 正确 True
- ✓ 错误 False

代码中可能有未记录的需求。

There maybe undocumented requirements in the code.

## Question 3

1 / 1 pts

请根据不同的控制流覆盖水平，选择正确的排序。

Please choose the correct order in terms of the power of the different levels of control flow coverage.

- 语句覆盖 < 决策覆盖 < 复合条件覆盖 < 决策/条件覆盖 Statement coverage < decision coverage < multiple condition coverage < decision/condition coverage
- 决策覆盖 < 语句覆盖 < 决策/条件覆盖 < 复合条件覆盖 Decision coverage < Statement coverage < decision/condition coverage < multiple condition coverage
- ✓ 语句覆盖 < 决策覆盖 < 决策/条件覆盖 < 复合条件覆盖 Statement coverage < decision coverage < decision/condition coverage < multiple condition coverage

  决策覆盖总能满足语句覆盖。决策/条件覆盖总能满足决策覆盖和语句覆盖。复合条件覆盖总能满足另外三种覆盖。

  Decision coverage always satisfies statement coverage. Decision /condition always satisfies decision coverage and statement coverage. Multiple condition coverage always satisfies the other three.

- 语句覆盖 < 决策/条件覆盖 < 决策覆盖 < 复合条件覆盖 Statement coverage < decision/condition coverage < decision coverage < multiple condition coverage

Correct answer

## Question 4

1 / 1 pts

根据下方给定的代码，哪组测试用例可达到 100% 的语句覆盖率？

Given the code below, which set of test cases will achieve 100% statement coverage?

```
If a < 12 and b = 5 or c > 15
    X = 50;
Else
    X = 25;
If X = 25
    Z = 12;
Else
    Z = 15;
```

- ✓ 测试用例 Test Case 1: A = 6, B=5, C=17, X=50

  测试用例 Test Case 2: A=15, B=3, C=8, X=25

  每条语句都由两个测试用例所覆盖。

  Each statement is covered with 2 test cases.

- A=15, B=3, C=8, X=25
- A = 11, B=5, C=20, X=50
- 测试用例 1：A = 6，B=5，C=17，X=50 Test Case 1: A = 6, B=5, C=17, X=50

Correct answer

## Question 5

1 / 1 pts

根据下方给定的代码，需要多少测试用例才能达到 100% 的决策条件覆盖？

Given the code below, how many test cases are needed to achieve 100% decision condition coverage?

If a < 12 and b = 5 or c > 15

X = 50;

Else

X = 25;

- 8
- 3
- ✓ 2

我们需要测试每种条件的正确/错误（T/F）结果。采用2个测试用例即可完成，让每个条件正确或错误时，分别作为一个测试用例。

We need to test the T/F outcome of each condition. This can be completed in 2 test cases, with each condition being true as a test case and each condition being false as a test case.

- 6

## Question 6

1 / 1 pts

根据下方给定的代码，哪组测试用例可达到 100% 的决策覆盖？

Given the code below, which set of test cases will achieve 100% decision coverage?

If a < 12 and b = 5 or c > 15

X = 50;

Else

X = 25;

If X = 25

Z = 12;

Else

Z = 15;

- A=15, B=3, C=8, X=25
- 测试用例 Test Case 1: A = 6, B=5, C=17, X=50
- ✓ 测试用例 Test Case 1: A = 6, B=5, C=17, X=50
  测试用例 Test Case 2: A=15, B=3, C=8, X=25

第一个测试用例覆盖了“决策 1 正确”和“决策 2 错误”。第二个测试用例覆盖了“决策 1 错误”和“决策 2 正确”。

The first test case covers Decision 1 True and Decision 2 F. The second test case covers Decision 1 False and Decision 2 True.

- A = 11, B=5, C=20, X=50

# 模組1_講座2：結構測試

# 模組 1_講座2：知識檢查

Correct answer

## Question 1

1 / 1 pts

真假判断题

基本路径数量是指针对每种可能路径创建测试用例和线性组合所需的最小路径数量。

True or False?

The number of basis paths is the minimum number of paths needed to build test cases and linear combinations for every other possible path.

○ 错误 False

✓ ◉ 正确 True

这是基本路径的定义。

This is the definition of basis paths.

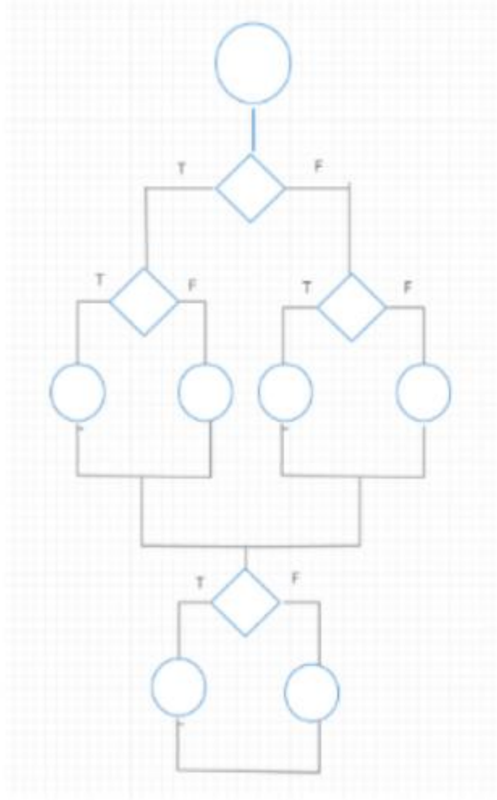
Correct answer

## Question 2

1 / 1 pts

TF_T, F_TT, TT_T, F_FT 和 TF_F，为以下控制流图可能的基本路径。

TF_T, F_TT, TT_T, F_FT, and TF_F are possible basis paths for the given control flow diagram.



○ 错误 False

✓ ◉ 正确 True

我们需要测试每个谓词的真和假。 以下是一种方法：

(1) 开始一条随机路径TF_T，其中 "_" 表示该位置的谓词无法执行。

(2) 翻转第一个（最上一个）决策得到: F_TT。 随机选择第三个值作为 T。

现在，完成测试第一个条件的真假。

(3) 重置回 (1) 并翻转第二个：TT_T。

现在，完成测试第二个条件的真假。

(4) 重置回 (2) 并翻转第三个：F_FT。

现在，完成测试第三个条件的真假。

(5) 重置回 (1) 并翻转第四个：TF_F。

现在，测试第四个条件的真假。

请注意，第 (5) 步您可以从 (4) 继续并翻转第四个条件。 它也可以覆盖第四个条件的错误条件。

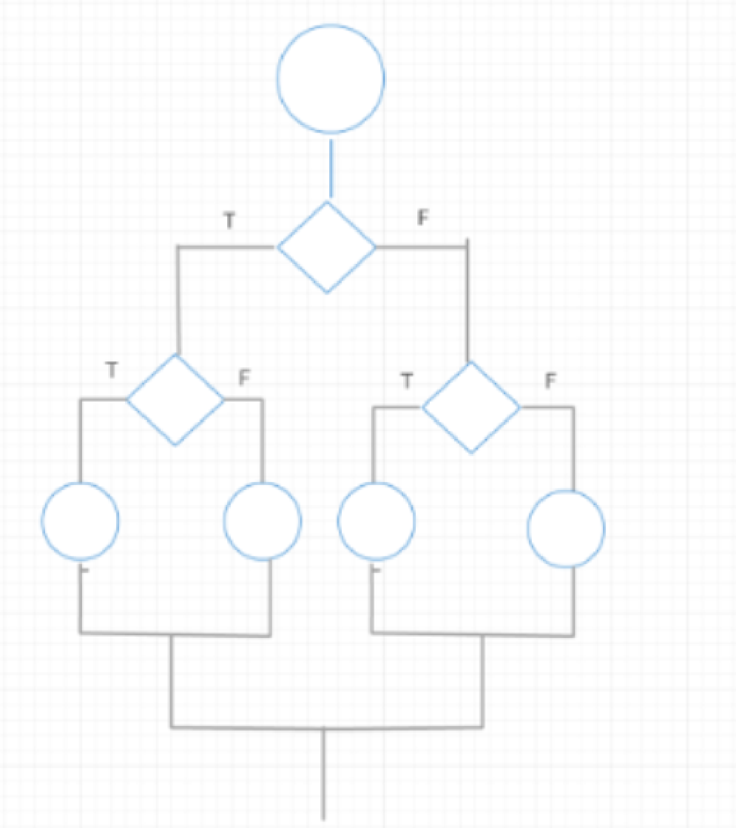
Correct answer

## Question 3

1 / 1 pts

给定控制流图的圈复杂度是多少？

What is the cyclomatic complexity of the given control flow diagram??




- 3
- 1
- ✓ 4

共有 3 个测试谓词 ，计算公式为测试谓词的数量 + 1。

There are three test predicates and the formula is # of test predicates + 1.

- 7

# 模組1_講座3：數據流測試

# 模組 1_講座3：知識檢查

Correct answer

## Question 1

1 / 1 pts

定义使用路径并不总是沿着定义清除路径。

A definition use path is not always along a definition clear path.

- 正确 True
- 错误 False ✓

我们想要沿着一条路径测试每个变量，在该路径中，每个变量的值都保持不变（如定义清除路径）

We want to test each variable along a path where it stays the same value through (i.e. a definition clear path)

Correct answer

## Question 4

1 / 1 pts

哪个不是定义使用覆盖中的变量集？

What is not a variable set used  in definition use coverage?

- C-Use(i)
- Def(i)
- P-Use(i)
- S-Use(i) ✓

这个不是定义使用覆盖中的变量集。

This is not a set of variables in definition use coverage.

Correct answer

## Question 2

1 / 1 pts

根据下方给定的代码，哪个是 正确的 DU 路径集？

Given the code below, what is the correct set of DU Paths?

```
Get a, b
x = 0          } Node 1 节点 1

If a < 5 (Predicate I) (谓词 i)
        Then c = x + 3 (Node 2)（节点2）
        Else c = a + 2 (Node 3)（节点3）

If b > 4 (Predicate II) (谓词 II)
        Then y = c + 4 (Node 4)（节点4）
        Else x = c + 2 (Node 5)（节点5）
```

Def1(a) = USEI(a) || USE3(a)

Def1(b) = USEII(b)

Def1(x) = USE2(x)

Def2(c) = USE4(c)

Def3(c) = USE4(c)

---

✓

Def1(a) = USEI(a) || USE3(a)

Def1(b) = USEII(b)

Def1(x) = USE2(x)

Def2(c) = USE4(c) || USE5(c)

Def3(c) = USE4(c) || USE5(c)

代码中定义了 5 个变量，且每个变量都以正确的使用方式展示。

There are 5 variables defined in the code, and each variable is displayed with the correct use.

---

Def1(a) = USE3(a)

Def1(b) = USEII(b)

Def1(x) = USE2(x)

Def2(c) = USE4(c) || USE5(c)

Def3(c) = USE4(c) || USE5(c)

---

Def1(a) = USEI(a) || USE3(a)

Def1(b) = USEII(b)

Def1(x) = USE2(x)

Def2(c) = USE4(c) || USE5(c)

Correct answer

## Question 3

1 / 1 pts

根据问题 3 中给定的代码和 DU 路径，以下测试用例可提供的覆盖率是多少？

Based on the code and the DU paths determined in Q3, what coverage do the test cases below provide?

A = 3; B=2

A=6, B=5

- 5/8
- 7/8
- ✓ 6/8

  覆盖的测试用例以粗体显示

  Covered test cases bolded

  Def1(a) = **USEI(a)** || **USE3(a)**

  Def1(b) = **USEII(b)**

  Def1(x) = **USE2(x)**

  Def2(c) = USE4(c) || **USE5(c)**

  Def3(c) = **USE4(c)** || USE5(c)
- 4/8

# 模組2：靜態分析

# 模組 2：知識檢查

Correct answer

## Question 1

1 / 1 pts

哪项不是数据流异常的示例？

What is not an example of a data flow anomaly?

- ✓ 定义并引用了变量 Variable is defined then referenced

  这不是异常。例如 x = 5; x = x + 3。

  This is not an anomaly. For example x = 5; x = x + 3.

- 定义变量，但从不使用它 Defining a variable but never using it
- 引用了未定义的变量 Referencing an undefined variable
- 定义并重新定义了变量，但未引用 Variable defined then redefined without being referenced

Correct answer

## Question 2

1 / 1 pts

静态分析通过查看在何处定义和使用变量来对程序中的数据流进行建模。

Static analysis models the flow of the data in a program by looking at where variables are defined and used.

- ✓ 正确 True

  这是静态分析的定义。

  This is the definition of static analysis.

- 错误 False

# 單元 4：測驗

Correct answer

**Question 1** **1 / 1 pts**

正确还是错误？

我们可以从控制流和数据流的角度对代码覆盖率进行评估。

True or False?

Code coverage can be assessed in terms of control flow and data flow.

- 错误 False
- ✓ 正确 True

Correct answer

**Question 2** **1 / 1 pts**

正确还是错误？

结构测试提供了一种测试路径子集的方法。

True or False?

Structured testing provides a strategy for testing a subset of paths.

- 错误 False
- ✓ 正确 True

Correct answer

## Question 3

1 / 1 pts

正确还是错误？

符号执行可用于确定程序中的特定路径无法执行。

True or False?

Symbolic execution can be used to determine that a specific path in the program cannot be executed.

- ✓ 正确 True
- 错误 False

Correct answer

## Question 4

1 / 1 pts

正确还是错误？

语句覆盖总能满足决策覆盖。

True or False?

Statement coverage always satisfies decision coverage.

- 正确 True
- ✓ 错误 False

Wrong answer

## Question 5

0 / 1 pts

正确还是错误？

数据流定义使用DU覆盖查看每个定义的变量至少在一个可以使用的地方被测试。

True or False?

Data flow definition use coverage looks to see that every variable defined is tested in at least one of the places that it can be used.

- 正确 True
- 错误 False

Correct answer

## Question 6

1 / 1 pts

正确还是错误？

黄氏定理有助于减少异常测试期间路径上的迭代次数。

True or False?

Huang’s Theorem helps reduce the number of iterations over a path during anomaly testing.

- 正确 True
- 错误 False

Correct answer

## Question 7

**1 / 1 pts**

根据下方给定的代码，哪组测试用例可达到 100% 的语句覆盖率？

Given the code below, which set of test cases will achieve 100% statement coverage?

```
Input: a, b, c, X;
If a < 5 or b > 7
    X = 50;
    c = a + b;
Else
    X = 25;
    c = a − b;
If X = 50 and c > 6
    Z = 10;
Else
    Z = 12;
```

- 测试用例 1 Test Case 1: a = 3, b=10, c=13, X=50
  测试用例 2 Test Case 2: a=3, b=4, c=4, X=25
- 测试用例 1 Test Case 1: a = 3, b=10, c=13, X=50
  测试用例2 Test Case 2: a=7, b=8, c=15, X=50
- ✓ 测试用例 1 Test Case 1: a = 3, b=10, c=13, X=50
  测试用例 2 Test Case 2: a=5, b=1, c=4, X=25
- 测试用例 1 Test Case 1: a = 2, b=10, c=12, X=25
  测试用例 2 Test Case 2: a=3, b=4, c=4, X=25

Correct answer

## Question 8

1 / 1 pts

根据下方给定的代码，哪个是 定义使用DU 路径正确的变量集？

Given the code below, what is the correct set of DU Paths?

```
Get a, b
X = 0        } Node 1

If a >= 5 (Predicate I)
        Then c = x + 3 (Node 2)
        Else c = 0 (Node 3)

If b < 4 (Predicate II)
        Then b = c + 4 (Node 4)
        Else b = c + 2 (Node 5)
```

- ✓ Def1(a) = USEI(a)
  Def1(b) = USEII(b)
  Def1(x) = USE2(x)
  Def2(c) = USE4(c) || USE5(c)
  Def3(c) = USE4(c) || USE5(c)

- Def1(a) = USEI(a)
  Def1(b) = USEII(b)
  Def1(x) = USE2(x)
  Def2(c) = USE4(c)
  Def3(c) = USE4(c)

- Def1(a) = USEI(a)
  Def1(b) = USEII(b)
  Def1(x) = USE2(x)
  Def2(c) = USE4(c) || USE5(c)

- Def1(a) = USEI(a)
  Def1(b) = USEII(b)
  Def2(c) = USE4(c) || USE5(c)
  Def3(c) = USE4(c) || USE5(c)

Correct answer

## Question 9

1 / 1 pts

根据下面的代码，Z 的最终符号值是多少？

Given the code below, what is the final symbolic value of Z?

(0) Input X, Y, Z

(1) Y = 2*X + Z

(2) W = Y + X

(3) Z = W + Y

- $3X_0 + 2Z_0$
- $5X_0 + Z_0$
- ✓ $5X_0 + 2Z_0$
- $3X_0 + Z_0$

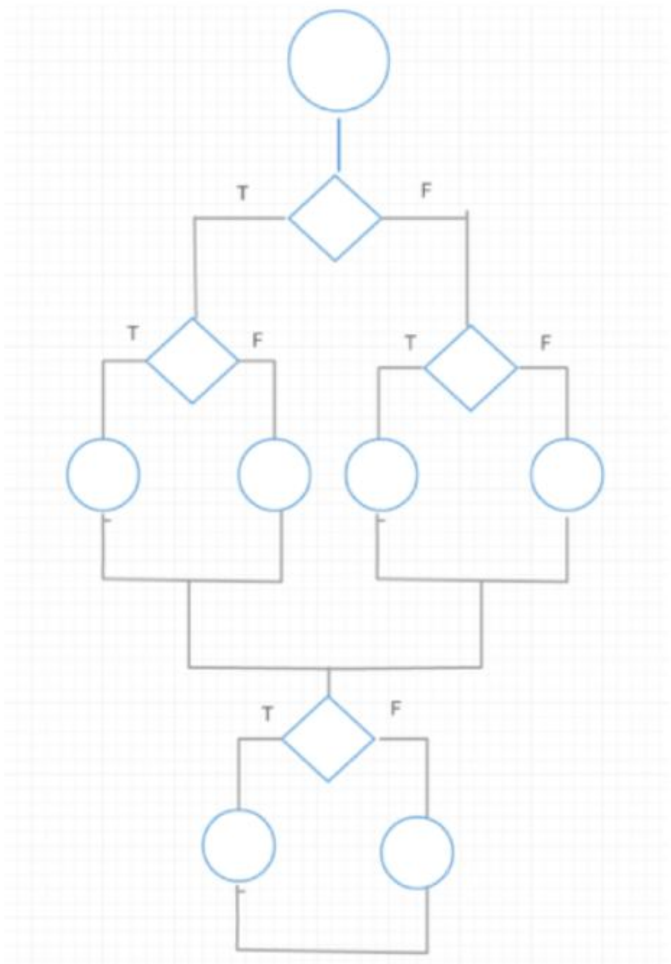
Correct answer

## Question 10

1 / 1 pts

给定控制流图的圈型复杂度是多少？

What is the cyclomatic complexity of the given control flow diagram?




- ✓ 5
- 4
- 6
- 7
- 8

# 單元五

# 模組1：性能測試

# 模組 1：知識檢查

Correct answer

**Question 1** 1 / 1 pts

正确还是错误？
性能测试涉及改变系统负载，并将结果与性能要求进行比对。

True or False?

Performance testing involves varying the load on the system and comparing the results against the performance requirements.

✓ 正确 True

必须对各种负载指定响应时间。The response time must be specified under various loads.

错误 False

Correct answer

**Question 3** 1 / 1 pts

正确还是错误？
分析资源使用情况有助于确定性能问题的潜在根源。

True or False?

Analysis of resource usage helps identify potential sources of performance issues.

✓ 正确 True

了解资源的使用方式/影响有助于通过查看关系确定性能瓶颈。Seeing how resources are used/affected helps identify performance bottlenecks through looking at relationships.

错误 False

Correct answer

## Question 2

1 / 1 pts

如何改善以下性能要求？
在 Canvas 上，提交作业后，确认界面将快速加载。

How can the following performance requirement be improved?

On Canvas, once an assignment is submitted, the confirmation screen should load quickly.

- 正常情况下，100 名学生同时在 Canvas 上提交作业，确认界面的加载时间应少于 2 秒。Under normal conditions of 100 students submitting an assignment at the same time on Canvas, the confirmation screen should load in less than 2 seconds.

  此要求说明了提交作业时的正常情况，并对"迅速"给出定量测定指标。This requirement states what the normal conditions are, when the assignment is submitted and gives quantitative measurements for "quickly".

- 正常情况下，100 名学生同时在 Canvas 上提交作业，确认界面将快速加载。Under normal conditions of 100 students submitting an assignment at the same time on Canvas, the confirmation screen should load quickly.

- 正常情况下，同时在 Canvas 上提交作业，确认界面的加载时间应少于 2 秒。When normal conditions of submitting an assignment at the same time on Canvas, the confirmation screen should load in less than 2 seconds.

- 正常情况下，100 名学生在 Canvas 上提交作业，确认界面的加载时间应少于 2 秒。Under normal conditions of 100 students submitting an assignment on Canvas, the confirmation screen should load in less than 2 seconds.

# 模組2：壓力測試

# 模組 2：知識檢查

Correct answer

**Question 1** 1 / 1 pts

下列哪项不是压力测试步骤？

Which of the following is not a stress testing step?

- 确定使资源饱和的方法 Determine an approach for saturating the resources
- 衡量并验证实际上是否已产生压力。Measure and verify that stress is actually achieved.
- ✓ 在没有产生压力的情况下运行额外的测试 Run additional tests where stress is not generated

  额外的测试是好的，但不是压力测试的一部分。Additional testing is good but is not a part of stress testing.
- 确定要产生压力的资源 Identify the resources to stress

Correct answer

**Question 2** 1 / 1 pts

正确还是错误？
应该在项目的最后几周计划压力测试。

True or False?

Stress testing should be scheduled during the last couple of weeks of the project.

- 正确 True
- ✓ 错误 False

  压力测试应尽早开始。Stress testing should begin earlier.

# 模組3：容量測試

# 模組 3：知識檢查

Correct answer

**Question 1** 1 / 1 pts

正确还是错误？
容量测试旨在验证系统在短时间内进行大量活动时是否满足其需求。

True or False?

Volume testing looks to verify that a system meets its requirements when it is subjected to a large volume of activity over a short amount of time.

✓ 错误 False

容量测试查看长时间内的活动。Volume testing is looking at activity over an extended period of time.

正确 True

Correct answer

**Question 2** 1 / 1 pts

下列哪项不是容量测试可找到的错误？

What is not an error targeted by volume testing?

- 资源耗竭 Resource depletion
- 计数器溢出 Counter overflow
- 内存泄漏 Memory leaks
- ✓ 逻辑错误 logic errors

功能测试技术可找到逻辑错误。Functional testing techniques target logic errors.

# 模組4：配置測試

# 模組 4：知識檢查

Correct answer

**Question 1** 1 / 1 pts

以下哪项不是配置测试的策略？

Which of the following is not a strategy for configuration testing?

- ✓ 对所有可能的配置执行测试 Execute tests on all possible configurations

  这通常无法做到，也没有必要。This is normally impossible and not necessary.

- 根据故障风险选择要测试的配置。Select configurations to test based on risk of failure.

- 利用试验设计确定要测试的配置。Utilize design of experiments to identify configurations to test.

- 测试最高配置和最低配置。Test maximum and minimum configurations.

Correct answer

**Question 2** 1 / 1 pts

正确还是错误？
配置测试旨在验证系统的不同配置是否满足系统的功能和性能要求。

True or False?

Configuration testing looks to verify that the functional and performance requirements of a system are met for different configurations of the system.

- 错误 False

- ✓ 正确 True

  这是配置测试的定义。This is the definition of configuration testing.

# 模組5：回歸測試

# 模組 5：知識檢查

Correct answer

## Question 1

1 / 1 pts

正确还是错误？
波动效应分析要求开发人员确定更改对其他需求的影响。

True or False?

Ripple effect analysis requires developers to identify the impact of changes on other requirements.

- 错误 False
- 正确 True

这是波动效应分析的定义。This is the definition of ripple effect analysis.

Correct answer

## Question 2

1 / 1 pts

下列哪项不是可以在代码中引入错误的原因？

Which of the following is not a reason why errors can be introduced in code?

- 代码波动效应 Code ripple effects
- 都不是 None of these

所有这些原因都是可以在代码中引入错误的方式。All of the reasons are ways that errors can be introduced in code.

- 非预期功能交互 Unintended feature interactions
- 代码更改引起了性能更改 Changes in performance caused by code changes.

Correct answer

## Question 3

1 / 1 pts

正确还是错误？
回归测试旨在确保当前的软件更改适用于系统。

True or False?

Regression testing looks to ensure the current software changes work with the system.

正确 True

错误 False

回归测试旨在确保以前的软件仍然能够与当前添加的软件协同工作。Regression testing looks to ensure that previous software is still working with current software additions.

# 模組6：移動端測試 (無知識檢查)

# 單元 5：測驗

Correct answer

**Question 1** 1 / 1 pts

正确还是错误？

无论系统是否崩溃，压力测试都不需要分析程序输出。

True or False?

Stress testing does not require analysis of program outputs only whether or not the system crashes.

- 正确 True
- ✓ 错误 False

Correct answer

**Question 2** 1 / 1 pts

正确还是错误？

性能需求要说明负载变化如何影响响应时间。

True or False?

Performance requirements specify how load variation affects response time.

- ✓ 正确 True
- 错误 False

Wrong answer

**Question 3** 0 / 1 pts

正确还是错误？

基于最流行的客户配置的测试可最大限度地降低配置测试的风险。

True or False?

Testing based on most popular customer configurations minimizes risk in configuration testing.

- 正确 True
- 错误 False

Correct answer

**Question 4** 1 / 1 pts

正确还是错误？

回归测试仅在系统层次上执行。

True or False?

Regression testing is only performed at the system level.

- 正确 True
- 错误 False

Correct answer

## Question 5

1 / 1 pts

正确还是错误？

内存泄漏是指当测试负载太大时导致内存损坏而减少内存容量。

True or False?

Memory leak refers to memory damage and size reduction because of high testing load.

- 错误 False
- 正确 True

## Question 6

0.5 / 1 pts

在定义一项性能需求时，哪些因素必须定量说明？有多个正确答案。

When defining a performance requirement, which factor must be included? There are multiple correct answers.

- [x] 反应时间。Response time.
- [x] 负载。Load.
- [x] 可靠性。Reliability.
- [ ] 测试用例。Test case.

Correct answer

## Question 7

1 / 1 pts

下列哪项不是性能测试的入口准则？

Which is not an entry criterion for performance testing?

- 存在代表性测试环境 Representative test environment exists
- 性能需求已定义并且可测试 Performance requirements are defined and testable
- 稳定的运行系统 Stable system that is working
- ✓ 功能测试已完成 Functional testing is complete

Correct answer

## Question 8

1 / 1 pts

下列哪项不是创建配置组合以在配置测试中进行测试的好方法？

Which is not a good way to create configuration combinations to test in configuration testing?

- DOE 结对组合测试 DOE Pairwise Combination Testing
- 基于风险的测试 Risk based testing
- ✓ 随机测试 Randomized testing
- 边界值测试 Boundary value testing

Correct answer

## Question 9

1 / 1 pts

选择所有正确的陈述。有多个正确答案。

Choose all correct statements. There are multiple correct answers.

- [x] 压力测试通常在短时间内执行。Stress testing is usually performed in a short period of time.
- [ ] 压力测试通常在长时间内执行。Stress testing is usually performed over a long period of time.
- [ ] 容量测试通常在短时间内执行。Volume testing is usually performed in a short period of time.
- [x] 容量测试通常在长时间内执行。Volume testing is usually performed over a long period of time.

Correct answer

## Question 10

1 / 1 pts

哪种测试方法通常用来发现内存泄漏？

What testing method is usually used for detecting Memory leak?

- ( ) 回归测试
- (•) 容量测试
- ( ) 配置测试
- ( ) 压力测试

# 單元六

# 模組1：錯誤檢測、恢復和服務能力測試

# 模組 1：知識檢查

Correct answer

**Question 1** 1 / 1 pts

下列哪项活动通常不是验证服务能力需求的一部分?

Which of the following activities is not normally a part of verifying a serviceability requirement?

- 问题纠正 Problem correction
- ✓ 实现错误恢复代码 Implement error recovery code

  实现错误恢复代码是开发过程的一部分，不是验证的一部分。Implementing error recovery code is part of the development process and not part of verification.

- 问题验证 Problem verification
- 问题隔离 Problem isolation

Correct answer

**Question 2** 1 / 1 pts

正确还是错误？

系统需要检测所有可能的失效并从失效中恢复。

True or False?

A system needs to detect and recover from all possible failures.

- ✓ 错误 False

  系统必须能够检测指定的失效并从失效中恢复。A system must be able to detect and recover from specified failures.

- 正确 True

# 模組2：易用性測試

# 模組 2：知識檢查

Correct answer

**Question 1** 1 / 1 pts

正确还是错误？
易用性测试还应该可靠并有效。

True or False?

Usability tests should also be reliable and valid.

- 错误 False
- ✓ 正确 True

测试应该确保结果可以复制，并且测试相关的内容。
Tests should ensure that the results can be replicated and are testing something of relevance.

Correct answer

**Question 2** 1 / 1 pts

下列哪项不是易用性测试的目标？

What is not an objective of usability testing?

- 确保用户可以按预期执行任务 Ensuring that the user can perform tasks as intended
- ✓ 确保用户不能访问他们无权访问的功能 Ensuring that the user does not access functions for which they do not have permission..

这不是易用性测试的一部分。这是安全性测试的一部分。This is not a part of usability testing. This would be part of security testing.

- 确保用户对界面满意 Ensuring that the user is satisfied by the interface
- 确保用户不受可能操作的影响 Ensuring that the user is protected from possible actions

Correct answer

## Question 3

1 / 1 pts

正确还是错误？
易用性测试应在集成测试之后立即执行。

True or False?

Usability testing should be performed immediately after integration testing.

- 错误 False

  易用性测试应在原型设计阶段的早期开始。Usability testing should begin early on during the prototyping phase.

- 正确 True

# 模組3_講座1：可靠性測試

# 模組 3_講座1：知識檢查

Correct answer

## Question 1

1 / 1 pts

正确还是错误？
“系统很有可能在接下来的十分钟内失效。”是可靠性需求的一个示例。

True or False?

“System failure during the next ten minutes is very likely.” is an example of a reliability requirement.

- ✓ 错误 False

  可靠性需求必须用概率表示。“很可能”不是概率。Reliability requirements must be expressed in terms of probabilities. “Very likely” is not a probability.

- 正确 True

Correct answer

## Question 2

1 / 1 pts

下列哪项不是获取运行概况数据的方法？

Which of the following is not a method for obtaining operational profile data?

- 竞争系统 Competing systems
- ✓ 测试人员意见 Tester opinions

  测试人员意见可能无法代表使用该系统的用户，并且会提供不正确的运行概况数据。Tester opinions may not be representative of people using the system and would provide incorrect operational profile data.

- 市场营销 Marketing
- 现有系统 Existing systems

Correct answer

## Question 3

1 / 1 pts

5 个 9 可用性要求每年最多有多少分钟不可用？

Five nines availability requires a maximum of how many minutes of unavailability per year?

- 200 分钟 200 minutes
- 999 分钟 999 minutes
- ✓ 5 分钟 5 Minutes

  需求是 5 分钟。5 minutes is the requirement.

- 10分钟 10 Minutes

# 模組3_講座2：可靠性模型

# 模組 3_講座2：知識檢查

Correct answer

## Question 1

1 / 1 pts

正确还是错误？
软件可靠性增长模型利用历史数据预测系统可靠性。

True or False?

Software Reliability Growth models utilize historic data to predict system reliability.

- ✓ 错误 False

  数据是从系统测试中获得的，而不是来自历史数据。
  Data is obtained from system testing, not historic data.

- 正确 True

Correct answer

## Question 2

1 / 1 pts

下列哪项不是可靠性模型作出的典型假设？

Which of the following is not a typical assumption made by reliability models?

- 模型由数学分布表示。Models are represented by mathematical distributions.
- 修复不会引入新的错误。No new errors are introduced by fixes.
- ✓ 失效强度是恒定的。Failure intensity is constant.

  模型通常假设随着时间的推移，可靠性将随着缺陷的发现和消除而提高。Models typically assume reliability will improve over time as defects are found and removed.

- 始终使用运行轮廓。Operational profiles are always used.

Correct answer

## Question 3

1 / 1 pts

正确还是错误？
可靠性模型有助于回答问题：“何时停止测试？”。

True or False?

Reliability models help to answer the question: “when do we stop testing?”.

- 错误 False
- ✓ 正确 True

模型通常显示可靠性如何随时间发生变化以及我们的失效强度是多少。Models typically show how reliability changes over time and what our failure intensity is.

# 模組4：安全性測試

# 模組 4：知識檢查

Correct answer

## Question 1

1 / 1 pts

下列哪项不需要在安全性测试期间加以考虑？

Which of the following does not need to be considered during security testing?

- 可用性 Availability
- 保密性 Confidentiality
- 完整性 Integrity
- 可靠性 Reliabliity

可靠性和安全性不是一回事。 系统可能安全，但不可靠，反之亦然。Reliability and security are not the same.  A system can be secure and not reliable and vice versa.

Correct answer

## Question 2

1 / 1 pts

正确还是错误？

对内存不足的系统进行压力测试可能会导致安全风险。

True or False?

Stress testing a system with low memory could result in a security risk.

- 正确 True

内存不足的压力测试可能会使数据不受保护。Stress testing with low memory could leave data unprotected.

- 错误 False

Correct answer

## Question 3

1 / 1 pts

正确还是错误？
安全性测试将检查是否保护私有数据免受未经授权用户的访问。

True or False?

Security testing checks to see if private data is protected from unauthorized users.

- [x] 正确 True

这是安全性测试的定义。This is the definition of security testing.

- [ ] 错误 False

# 單元 6：測驗

Correct answer

**Question 1** 1 / 1 pts

正确还是错误？True or False?

使用用例有助于开发定量的和可测量的易用性测试。

Use cases can help with developing quantitative and measurable usability tests.

- ✓ True
- False

Correct answer

**Question 2** 1 / 1 pts

正确还是错误？True or False?

运行轮廓可用于性能分析。

Operational profiles can be used in performance analysis.

- ✓ True
- False

Correct answer

## Question 3

1 / 1 pts

正确还是错误？True or False?

服务能力测试的目标是验证是否满足服务能力需求。

The objective of serviceability testing is to verify that serviceability requirements are being met.

- ✓ True
- False

Correct answer

## Question 4

1 / 1 pts

正确还是错误？True or False?

认证测试的目标是消除导致失效的故障。

The goal of certification testing is to remove faults that have caused failures.

- True
- ✓ False

Correct answer

## Question 5

1 / 1 pts

正确还是错误？True or False?

即使系统的可靠性低，它也可以具有高可用性。

A system can have high availability even if it has low reliability.

- ✓ ◉ True
- ○ False

Correct answer

## Question 6

1 / 1 pts

下列哪项对在系统中实现高可靠性和可用性没有帮助？

Which of the following does not help achieve high reliability and availability in a system?

- ○ 实现缺陷预防技术 Implementing defect prevention techniques
- ○ 指定需要检测哪些失效 Specifying what failures need to be detected
- ✓ ◉ 验证安全性需求 Verifying security requirements
- ○ 实现故障容错设计和代码 Implementing fault tolerant design and code

Correct answer

## Question 7

1 / 1 pts

下列哪些项是执行 GUI 安全性测试的方法？选择所有正确答案。

Which of the followings are methods to perform GUI security testing? Select all correct answers.

- [ ] 检查是否所有GUI 输入框都能实现数据输入。 Checking if all GUI textboxes can take data input.
- [x] 检查对 GUI 的可能有害输入 Checking potentially harmful inputs to the GUI
- [x] 检查访问数据的所有可能方式 Checking all the possible ways to access data
- [ ] 检查是否所有GUI 按键都能触发一项功能。 Checking if all GUI buttons can trigger a function.

Correct answer

## Question 8

1 / 1 pts

下列哪项不是开发和运行易用性测试的一部分？

Which of the following is not a part of developing and running a usability test?

- ( ) 试行测试 Piloting the test
- ( ) 确定测试用户 Identifying test users
- (•) 强调用户正在接受测试 Emphasizing that the user is being tested
- ( ) 创建合理的测试任务 Creating reasonable test tasks

Correct answer

## Question 9

1 / 1 pts

下列哪项是易用性测试过程中使用的评估类型？选择所有正确答案。

Which of the followings are a type of evaluation used during usability tests? Select all correct answers.

- [ ] 生成性评估 Generative evaluation
- [x] 形成性评估 Formative evaluation
- [x] 总结性评估 Summative evaluation
- [ ] 安全性评估 Security evaluation

Correct answer

## Question 10

1 / 1 pts

下列哪项不属于良好的易用性需求？

Which of the following is not included in a good usability requirement?

- ( ) 效率 Efficiency
- ( ) 易学性 Learnability
- (•) 系统准确性 System accuracy
- ( ) 可记忆性 Memorability

# 單元七

# 模組1：測試計畫概述

# 模組 1：知識檢查

Correct answer

## Question 1

1 / 1 pts

正确还是错误？
系统测试计划涉及了解测试目标和约束。

True or False?

System test planning involves both an understanding of the test objectives and the constraints.

错误 False

正确 True

测试计划必须考虑目标和约束。Test plans must address objectives and constraints.

Correct answer

## Question 2

1 / 1 pts

正确还是错误？
仅系统测试需要测试计划。

True or False?

Test planning is only needed for system testing.

错误 False

所有测试级别的测试计划应为书面形式。Test plans should be written for all testing levels.

正确 True

Correct answer

## Question 3

1 / 1 pts

下列哪项不是系统测试计划的一部分？

Which of the following is not a component of a system test plan?

- 测试策略 Test Strategy
- 测试环境的规格说明 Specification of the test environment
- 测试用例集及其预期结果 Set of test cases and their expected results

  测试用例不是测试计划的一部分。Test cases are not a part of the test plan

- 进度表 Schedule

# 模組2：測試進度表

# 模組 2：知識檢查

Correct answer

**Question 1** **1 / 1 pts**

正确还是错误？

鼓励参与者遵守进度表的一个好方法是为他们提供帮助制定进度表的机会。

True or False?

A good way to encourage participants to commit to a schedule is provide them with an opportunity to help develop the schedule.

错误 False

✓ 正确 True

有重要贡献可以鼓励参与者想要完成项目。

Having important contributions encourages participants to want to complete the project.

Correct answer

**Question 2** **1 / 1 pts**

正确还是错误？

PERT 图有助于记录测试计划中的相关性。

True or False?

A PERT chart helps document dependencies in a test schedule.

错误 False

✓ 正确 True

PERT 图旨在确定相关性。

PERT charts were designed to identify dependencies.

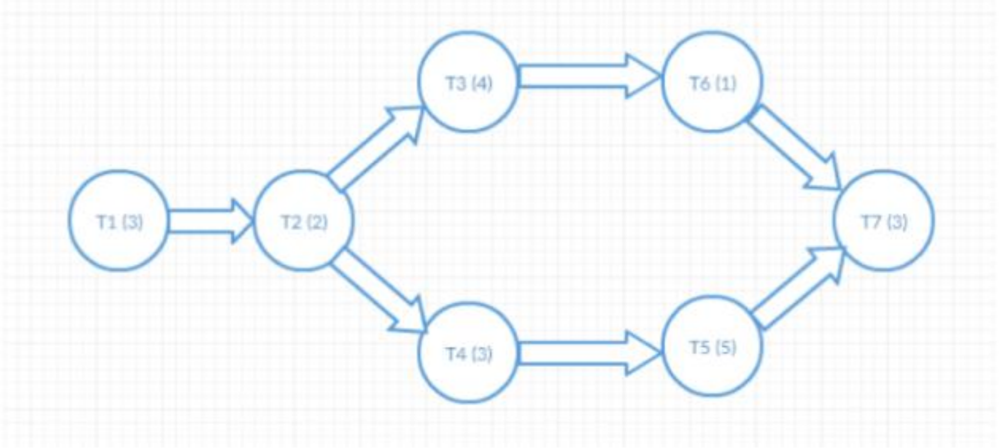
Correct answer

## Question 3

1 / 1 pts

根据下面的 PERT 图，仅有一个人员的情况下，完成此项目的最短时间是多少？

Given the following PERT chart, what is the minimum time to complete this project with only one person working?




- 17
- 21 ✓

如果仅由一个人完成，则必须将每个任务的时间相加。3 + 2 + 4 +3 + 1 + 5 + 3 = 21.

The time for each task must be added up if only completed by one person. 3 + 2 + 4 +3 + 1 + 5 + 3 = 21.

- 13
- 16

# 模組3_講座1：估算測試工作量

# 模組 3_講座1：知識檢查

Correct answer

**Question 1** 1 / 1 pts

正确还是错误？

系统的规模和复杂性将影响所需的测试时间。

True or False?

Size and complexity of a system affect the time needed to test it.

- 错误 False
- ✓ 正确 True

规模和复杂性是影响系统测试时间的两个因素。

Size and complexity are two factors that affect system testing time.

Correct answer

**Question 2** 1 / 1 pts

正确还是错误？

POFA（第一次尝试通过）预先指示了代码的质量。

True or False?

POFA (pass of first attempt) provides an early indication of the quality of code.

- 错误 False
- ✓ 正确 True

POFA 是第一次尝试时通过的测试百分比。

POFA is a percentage of tests that will pass on the first attempt.

Correct answer

## Question 3

1 / 1 pts

下列哪项不是开发测试估算的方法？

Which of the following is not an approach for developing test estimates?

- 使用历史数据 Using historical data
- ✓ 估算预期缺陷的数量 Estimating the number of expected defects

  即使人们知道预期缺陷的数量，也没有依据来估算检测缺陷的时间

  Even if one were to know the expected number of defects, there is no basis for estimating the time to detect them.

- 使用成本估算模型 Using a cost estimation model
- 使用开发估算的百分比 Using a percentage of the development estimate

# 模組3_講座2：測試估算步驟

# 模組 3_講座2：知識檢查

Correct answer

**Question 1** 1 / 1 pts

正确还是错误？

工作分解结构 (WBS) 有助于进行自底向上估算。

True or False?

A work breakdown structure (WBS) helps in bottom up estimation.

正确 True

WBS 有助于组织在自底向上估算过程中分解的部分。

A WBS helps to organize the parts that were broken down during bottom up estimation.

错误 False

Correct answer

**Question 2** 1 / 1 pts

正确还是错误？

工作量估算发生在分解部分之前和/或规模估算之后

True or False?

Effort estimation can take part before, concurrently, or after size estimation.

正确 True

错误 False

工作量基于项目的规模，仅在估算项目的规模之后才能确定。

Effort is based upon the size of the project, and can only be determined after the size of the project is estimated.

Correct answer

## Question 3

1 / 1 pts

下列哪项不是检查估算是否正确的策略？

Which of the following is not a strategy for checking if an estimate is correct?

- 检查产品类似部分的估算是否相同 Checking that estimates for similar parts of a product are the same
- 根据个人假设随机选择估算 Randomly selecting estimates based on personal assumptions

  这很可能不会提供有效的估算。

  This would most likely not provide a valid estimate.
- 与其他专家一起讨论 Discussing with other experts
- 查看历史数据 Looking at historical data

# 模組4：基於風險的測試

# 模組 4：知識檢查

Correct answer

## Question 1

1 / 1 pts

正确还是错误？

我们应尽早测试高风险区域，但无需详尽地测试这些区域。

True or False?

It is good to test high risk areas areas early, but they do not need to be tested thoroughly.

正确 True

错误 False

我们应对高风险区域进行更为详尽地测试

High risk areas should be tested more thoroughly.

Correct answer

## Question 2

1 / 1 pts

正确还是错误？

应根据风险暴露，对测试进行优先级排序。

True or False?

Testing should be prioritized based on the risk exposure.

错误 False

正确 True

风险暴露是指用不良事件发生的概率乘以该事件的后果。如数值较高，则应优先测试。

Risk exposure is the probability of an adverse event occurring multiplied by the consequences. If this value is high, it should be tested first.

# 模組5：測試出口準則

# 模組 5：知識檢查

Correct answer

**Question 1** 1 / 1 pts

正确还是错误？

使用历史数据来预测缺陷密度并不总是能够得到准确的预测。

True or False?

Using historical data to predict defect density may not always lead to accurate predictions.

- 错误 False
- ✓ 正确 True

当前项目的性能可能高于历史数据项目。

The current project may be performing better than the historical data projects.

Correct answer

**Question 2** 1 / 1 pts

根据下面的示例，估算还剩多少缺陷？

假设 X 组发现 20 个缺陷，Y 组发现 10 个缺陷。5 个缺陷是 X 组和 Y 组共有的。估算的剩余缺陷是多少？

Given the following example, how many defects are estimated to be remaining?

Assume Group X found 20 defects and Group Y found 10 defects. 5 defects were common to both Group X and Group Y. What is the estimated remaining defects?

- ✓ 15

(20+10)-5 = 25 个独特缺陷。(20*10)/5 = 40 个估算的总缺陷。估算的剩余缺陷等于估算的总缺陷减去独特缺陷。

(20+10)-5 = 25 unique defects. (20*10)/5 = 40 estimated total defects. Estimated remaining defects is estimated total defects – unique defects.

- 5
- 25
- 40

Correct answer

## Question 3

1 / 1 pts

正确还是错误？

缺陷植入通常与运行概况一起使用。

True or False?

Defect seeding is typically used with operational profiles.

- ✓ 错误 False

  缺陷池通常与运行概况一起使用。

  Defect pooling is typically used with operational profiles.

- 正确 True

# 模組6：測試文檔

# 模組 6：知識檢查

Correct answer

**Question 1** 1 / 1 pts

以下哪项不是测试事件报告的描述内容？

What is not included in a test incident report description?

- ✓ 缺陷的优先级 Priority of the defect

  缺陷的优先级不是由发现问题的测试人员设定的，而应由管理层做出判定。

  Defect priority is not set by the tester who finds the problem but is a management decision.

- 预期结果 Expected results
- 日期和时间 Date and time
- 重复操作 Attempts to repeat

Correct answer

**Question 2** 1 / 1 pts

正确还是错误？

严重程度反映的是错误对客户的影响。

True or False?

Severity reflects the customer impact of the error.

- 错误 False
- ✓ 正确 True

  这是严重程度的定义。

  This is the definition of severity.

Correct answer

## Question 3

1 / 1 pts

正确还是错误？

好的测试用例有其自身的标识，并能理想地映射到需求。

True or False?

Good test cases ideally map back to requirements and have their own identifiers.

- 错误 False
- ✓ 正确 True

映射回需求可确保，测试用例在测试输入和输出规格。标识有助于测试用例的追溯。

Mapping back to requirements ensures that the test cases are testing the input and output specifications. Identifiers help with traceability of test cases.

Correct answer

## Question 4

1 / 1 pts

以下哪项不是测试总结报告的内容？

Which of the following is not a part of a test summary report?

- 测试内容总结 Summary of what was tested
- ✓ 改善测试过程的建议 Recommendations for improving the testing process

这将是测试或项目事后回顾的一部分。

This would be a part of a testing post mortem or project retrospective.

- 综合评估 Comprehensive assessments
- 评价 Evaluation

# 單元 7：測驗

Correct answer

## Question 1

1 / 1 pts

正确还是错误？

敏捷方法需要遵循 IEEE 标准来编写测试文档。

True or False?

Agile methods require IEEE standards for test documentation to be followed.

- 正确 True
- ✓ 错误 False

IEEE standards were defined for traditional testing methods. However, agile testing could benefit from the standards. It is optional for agile methods. In the question, the word "require" means "must".

Correct answer

## Question 2

1 / 1 pts

正确还是错误？

高严重程度错误意味着高优先级错误。

True or False?

High severity errors imply high priority errors.

- 正确 True
- ✓ 错误 False

Correct answer

## Question 3

1 / 1 pts

正确还是错误？

当创建测试进度表时，不需要列出执行任务所需的工作量和资源。

True or False?

Having a list of effort and resources needed to perform tasks is not required when creating a test schedule.

- [x] 错误 False
- [ ] 正确 True

Correct answer

## Question 4

1 / 1 pts

正确还是错误？

风险管理有助于确定进度表中所需的应急时间。

True or False?

Risk management helps determine the amount of contingency time needed in a schedule.

- [x] 正确 True
- [ ] 错误 False

Correct answer

## Question 5

1 / 1 pts

正确还是错误？

被测程序的质量不会影响所需的测试时间。

True or False?

The quality of the program under test does not affect the time needed to test it.

- ✓ ◉ 错误 False
- ○ 正确 True

Correct answer

## Question 6

1 / 1 pts

系统测试计划中要涉及下列哪项？选择所有正确答案。

Which of the followings are addressed in a system test plan? Select all correct answers.

- ☐ 预期测试结果 Expected Test results
- ✓ ☑ 测试目标 Test objectives
- ✓ ☑ 进度表 Schedule
- ✓ ☑ 风险管理 Risk management

Correct answer

## Question 7

1 / 1 pts

下列哪因素是导致开发人员在软件应用程序中造成错误数增加的因素？选择所有正确答案。

Which of the followings are factors that contribute to the number of errors created by developers in a software application? Select all correct answers.

- [x] 代码复杂性 Code complexity
- [ ] 测试环境 The test environment
- [x] 代码更改的程度 Degree of code changes
- [x] 开发人员编写代码的经验 Experience of the developers working on the code

Correct answer

## Question 8

1 / 1 pts

下列哪项是导致估算不准确的原因？选择所有正确答案。

Which of the followings are reasons for inaccurate estimates? Select all correct answers.

- [x] 在开发估算时缺乏分析 Lack of analysis when developing estimates
- [x] 缺乏历史数据 Lack of historical data
- [ ] 人员流失 Losing personnel
- [x] 忽视任务 Overlooked tasks

## Question 9

0.67 / 1 pts

下列哪些是用于分析软件就绪情况的因素？选择所有正确答案。

Which of the followings are factors used to analyze software readiness? Select all correct answers.

- 软件中的剩余缺陷 Remaining defects in the software

  The remaining number of defects is unknow and cannot be used to analyze software readiness.

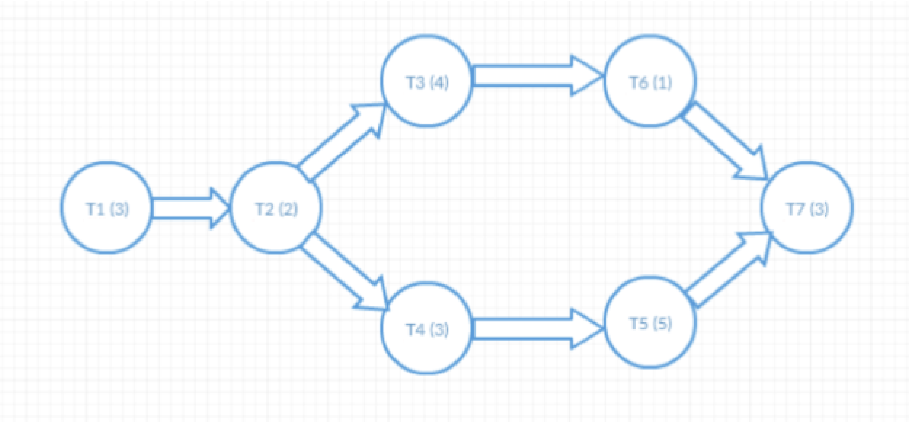
- 每时间单位的失效数 Number of failures per unit of time
- 失效平均时间 Mean time to failure
- 失效强度 Failure intensity

## Question 10

1 / 1 pts

已知下面的 PERT 图，如果只有一个人来完成此项目中的所有任务，最短时间是多少？

Given the following PERT chart, what is the minimum time to complete this project if there one person only to perform all the tasks?




- 21
- 16
- 17
- 13

# 單元八

# 模組1：測試追蹤

# 模組 1：知識檢查

Correct answer

**Question 1** 1 / 1 pts

以下哪项不是按照计划跟踪测试进度的方法？

Which of the following is not a method to track test progress against a plan?

- 查看已测试需求的百分比。Looking at the percentage of requirements tested
- 查看已执行测试的百分比。Looking at the percentage of tests executed
- 查看已开发测试的百分比。Looking at the percentage of tests developed
- ✓ 查看已通过测试的百分比。Looking at the percentage of tests that pass

识别没有按照计划显示进度的通过测试。Identifying tests that pass does not show progress against a plan.

Correct answer

**Question 2** 1 / 1 pts

根据以下挣值，项目是提前完成，按时完成，还是超时完成？

Given the following earned values, is the project ahead of, on, or behind schedule?

| | |
|---|---|
| BCWS | 150 |
| BCWP | 150 |
| ACWP | 100 |

- 超时完成 Behind schedule
- 提前完成 Ahead of schedule
- ✓ 按时完成 On schedule

BCWS 和 BCWP 等效。The BCWS and BCWP are equivalent.

Correct answer

## Question 3

1 / 1 pts

正确还是错误？
挣值主要用于标识进度差异。

True or False?

Earned values are used primarily to identify schedule variance.

- 正确 True
- 错误 False

同时跟踪进度和预算差异。Both schedule and budget variance are tracked.

# 模組2：測試過程改進

# 模組 2：知識檢查

Correct answer

## Question 1

1 / 1 pts

正确还是错误？
分析对于执行客观过程分析很重要。

True or False?

Analytics are important in performing objective process analysis.

- 错误 False
- ✓ 正确 True

分析可帮助你了解过程。Analytics provides visibility into the process.

Correct answer

## Question 2

1 / 1 pts

以下哪项不是目标问题度量 (GQM) 范例的一部分？

Which of the following is not a part of the Goal Question Metric (GQM) paradigm?

- 确定必须回答以达到目标的问题。Derive questions that must be answered to meet the goals
- 定义测量过程的目标。Define the goals of the measurement process
- ✓ 委派任务给团队成员，以达到设定的目标。Delegate tasks to team members to reach the goals defined

GQM 范例中不包含此项。This is not included in the GQM paradigm.

- 制定回答问题的度量。Develop metrics to answer the questions

Correct answer

## Question 3

1 / 1 pts

正确还是错误？
在组织中进行过程改进时，重点应放在检查“正式过程”上。

True or False?

When doing process improvement in an organization, the focus should be on examining the “official process”.

正确 True

错误 False

正式过程是你需要完成的内容。 过程改进始于检查“实际过程”。The official process is what you are supposed to do.  Process improvement begins with examining the “actual process”.

# 模組3：測試外包

# 模組 3：知識檢查

Correct answer

## Question 1

1 / 1 pts

正确还是错误？
外包测试始终比内部执行测试便宜。

True or False?

Outsourcing testing is always cheaper than performing the testing in-house.

错误 False

与外包计划和跟踪相关的成本。There is a cost associated with outsourcing planning and tracking.

正确 True

Correct answer

## Question 2

1 / 1 pts

正确还是错误？
工作验收标准由分包商指定。

True or False?

Acceptance of work criteria are specified by the subcontractor.

正确 True

错误 False

工作验收是正式程序，具有组织制定的合同性质。Acceptance of work is a formal procedure that has contractual implications developed by the organization.

Correct answer

## Question 3

1 / 1 pts

下列哪项不是外包协议的一部分？Which of the following is not a component of an outsourcing agreement?

- 技术规格 Technical specifications
- ✓ 负责项目的具体团队成员 Specific team members working on the project

  外包协议无需提供这种级别的详细信息。This level of specifics is not needed for an outsourcing agreement.

- 合同 Contract
- 工作说明书 Statement of work

Correct answer

## Question 4

1 / 1 pts

正确还是错误？
在制定分包商管理计划时，估算仅由分包商执行。

True or False?

When developing a subcontractor management plan, estimation is only performed by the subcontractor.

- 正确 True
- ✓ 错误 False

  公司或个人应首先估算所需资源，以确定分包商的估算是否合理。A company or individual should first estimate the resources needed to determine if a subcontractor estimate is reasonable.

# 模組4：人員管理

# 模組 4：知識檢查

Correct answer

**Question 1** 1 / 1 pts

正确还是错误？
系统测试负责人仅需与测试团队互动。

True or False?

A system test lead only has to interact with the test team.

- ✓ 错误 False

  他们还需要与开发团队和所有其他相关团队互动。They will also need to interact with the development team and any other teams involved.

- 正确 True

Correct answer

**Question 2** 1 / 1 pts

正确还是错误？
软件测试员的动力在于他们的成就得到认可。

True or False?

Software testers are motivated by recognition of their accomplishments.

- ✓ 正确 True

  被认可是最强大的动力之一。Recognition is one of the strongest motivators.

- 错误 False

Correct answer

## Question 3

1 / 1 pts

以下哪项可能不是软件测试员的主要动力？

Which of the following may not be a primary motivator for software testers?

- 贡献自豪感 Pride in contribution
- 成就自豪感 Pride in accomplishment
- 金钱 Money

  钱虽然重要，但可能不是人们的主要动力。Although important, money may not be a primary motivator for people.

- 工作自豪感 Pride in work

# 模組5：軟體審查

# 模組 5：知識檢查

Correct answer

**Question 1** 1 / 1 pts

以下哪项不是软件检查过程的一部分？

Which of the following is not part of the software inspection process?

- 优化检查表 Utilization of checklists
- ✓ 对发现的任何问题提出可能的修复方法 Bring in potential fixes for any problems found

  问题修复方法不应在审查时讨论。

  Fixes to problems should not be discussed in an inspection.
- 度量分析有助于改进过程 Metrics analysis to help process improvement
- 提前准备 Advanced preparation

Correct answer

**Question 2** 1 / 1 pts

正确还是错误？

培训团队是良好的测试文档审查的主要目标。

True or False?

The primary goal of a good test documentation inspection is to educate the team.

- ✓ 错误 False

  主要目标是发现缺陷。

  The primary goal is to find defects.
- 正确 True

Correct answer

## Question 3

1 / 1 pts

正确还是错误？

检查表可指导缺陷检测过程。

True or False?

Checklists guide the defect detection process.

- 错误 False
- 正确 True

检查表有助于系统地执行测试。

Checklists help perform testing systematically.

# 模組6：因果分析

# 模組 6：知識檢查

Correct answer

## Question 1

1 / 1 pts

正确还是错误？

沟通不畅或疏忽可能导致缺陷。

True or False?

Defects could be caused by miscommunication or oversight.

✓ 正确 True

这些是可能的根本原因。

These are possible root causes.

错误 False

Correct answer

## Question 2

1 / 1 pts

正确还是错误？

疏忽问题可以通过更好的工具和检查表减少。

True or False?

Oversight issues could be reduced with better tools and checklists.

错误 False

✓ 正确 True

疏忽问题可以通过更好的工具和检查表减少。

Tools and checklists can help detect oversight errors.

Correct answer

## Question 3

1 / 1 pts

正确还是错误？

原因分析试图识别缺陷的根本原因以及减少未来出现缺陷的方法。

True or False?

Causal analysis tries to identify the root cause of defects and ways to decrease future occurrences.

- ✓ 正确 True

  这是原因分析的定义。

  This is the definition of causal analysis.

- 错误 False

# 模組7：測試成熟度模型

# 模組 7：知識檢查

Correct answer

## Question 1

1 / 1 pts

正确还是错误？

如果公司处于 CMMI 等级 4，则它们将仅具有已定义并已记录的过程。

True or False?

If a company is at level 4 of CMMI, they will only have processes that are defined and documented.

- ✓ 错误 False

  此为等级 3。

  This is level 3.

- 正确 True

Correct answer

## Question 2

1 / 1 pts

正确还是错误？

TMMI 等级 4 开始注重度量。

True or False?

TMMI starts to look at metrics in level 4.

- 错误 False
- ✓ 正确 True

  等级 4 引入度量。

  Metrics are introduced in level 4.

# 單元 8：測驗

Correct answer

**Question 1** 1 / 1 pts

易测试性**不**包含哪些属性？选择所有正确答案。

What characteristics are **not** included in testability? Select all correct answers.

- 可见性 visibility
- ✓ 可维护性 maintainability
- ✓ 可用性 availability
- 可控性 controllability

Wrong answer

**Question 2** 0 / 1 pts

以下哪些项是过程改进的一个阶段？选择所有正确答案。

Which of the followings is a process improvement phase? Select all correct answers.

- → 过程重新设计 Process redesign
- ✗ 按照当前流程与成员交谈 Talk with members following the current process
- → 分析当前过程 Analyze the current process
- ✓ 描述目标过程 Characterize the target process

Correct answer

## Question 3

1 / 1 pts

下列哪项不是工作陈述的一部分？

Which of the following is not a component of a statement of work?

- 识别将要进行的所有任务 Identify all tasks to be performed
- 识别维护职责 Identify maintenance responsibilities
- 识别要遵循的相关过程 Identify relevant processes to be followed
- ✓ 已执行工作的预算成本 Budgeted cost of the work performed

Correct answer

## Question 4

1 / 1 pts

以下哪些项是"好团队"的特点？选择所有正确答案。

Which of the followings are characteristics of a "good team"? Select all correct answers.

- ✓ 如果有团队成员遇到困难，其他人自觉帮忙 If a team member is struggling, others automatically help
- ✓ 团队成员彼此非常了解 Team members know each other very well
- ✓ 团队互动轻松 Team's interactions are comfortable
- 从不发生冲突 Conflict never occurs

Correct answer

## Question 5

1 / 1 pts

根据以下任务和挣值，在第 1 周后，项目是高于预算，符合预算，还是低于预算？

Given the following tasks and earned values, is the project above, on, or below budget?

| | | | |
|---|---|---|---|
| 1A | 30 | 2A | 10 |
| 1B | 50 | 2B | 20 |
| 1C | 20 | | |

第 1 周，将要完成：1A、1B、2A

第 1 周，实际完成：1A、1B、1C，成本是 90

Week 1 To be Completed: 1A, 1B, 2A

Week 1 Actually Completed: 1A, 1B, 1C at a cost of 90

- 符合预算 On budget
- ✓ 低于预算 Below budget
- 高于预算 Above budget

Correct answer

## Question 6

1 / 1 pts

根据以下任务和挣值，在第 1 周后，项目的进展是超前，按时，还是滞后？

Given the following tasks and earned values, is the project ahead of, on, or behind schedule after week 1?

| | | | |
|---|---|---|---|
| 1A | 30 | 2A | 10 |
| 1B | 50 | 2B | 20 |
| 1C | 20 | | |

第 1 周，计划要完成：1A、1B、2A

第 1 周，实际完成：1A、1B、1C，成本是 90的

Week 1 To be Completed: 1A, 1B, 2A

Week 1 Actually Completed: 1A, 1B, 1C at a cost of 90

- 滞后于计划 Behind schedule
- 符合计划 On schedule
- ✓ 超前于计划 Ahead of schedule

Correct answer

## Question 7

1 / 1 pts

正确还是错误？True or False?

作为外包合同的一部分，外包机构可能要求遵循特定的测试过程。

As part of the outsourcing contract, an organization might require that particular testing processes be followed.

- 正确 True
- 错误 False

Correct answer

## Question 8

1 / 1 pts

正确还是错误？True or False?

TMMI 根据活动、任务和职责实现的各种目标和子目标来区分其等级。

TMMI separates its levels based on various goals and subgoals that are achieved via activities, tasks, and responsibilities.

- 错误 False
- 正确 True

Correct answer

## Question 9

1 / 1 pts

正确还是错误？True or False?

在测试审查期间，不需要软件开发人员。

Software developers are not needed during test inspections.

- 正确 True
- 错误 False ✓

Correct answer

## Question 10

1 / 1 pts

正确还是错误？True or False?

GQM 提倡项目应收集尽可能多的度量。

GQM advocates that a project should collect as many metrics as possible.

- 正确 True
- 错误 False ✓